

1. CEILING FAN CONTROLLER N.I.E.S. CONNECT AS REQUIRED.
2. TOP IN ATTIC N.I.E.S. CONNECT AS REQUIRED.
3. TIME CLOCK N.I.E.S. CONNECT AS REQUIRED.
4. SOLATUBE 120V/1N N.I.E.S. CONNECT AS REQUIRED.
5. SOLATUBE CONTROL N.I.E.S. CONNECT AS REQUIRED.
6. GROUP MOUNT DISCONNECT SWITCH ON STEEL CHANNELS.
SECURE STEEL CHANNELS TO STEEL POST.
7. DOOR CONTROLLER POWER SUPPLY N.I.E.S. MOUNT ABOVE
CEILING. CONNECT AS REQUIRED.
8. DOOR CONTROLLER N.I.E.S. CONNECT AS REQUIRED. PROVIDE
ALL NECESSARY WIRING.
9. DOOR CONTROL CONSOLE N.I.E.S. CONNECT AS REQUIRED.
PROVIDE ALL NECESSARY WIRING.
10. DOOR EXIT DEVICE N.I.E.S. CONNECT AS REQUIRED.
PROVIDE ALL NECESSARY WIRING.

Dry Creek Joint Elementary School District

IDENTIFICATION STAMP DIV. OF THE
STATE ARCHITECT OFFICE OF
REGULATION SERVICES

APPLICATION NO.
02-115456

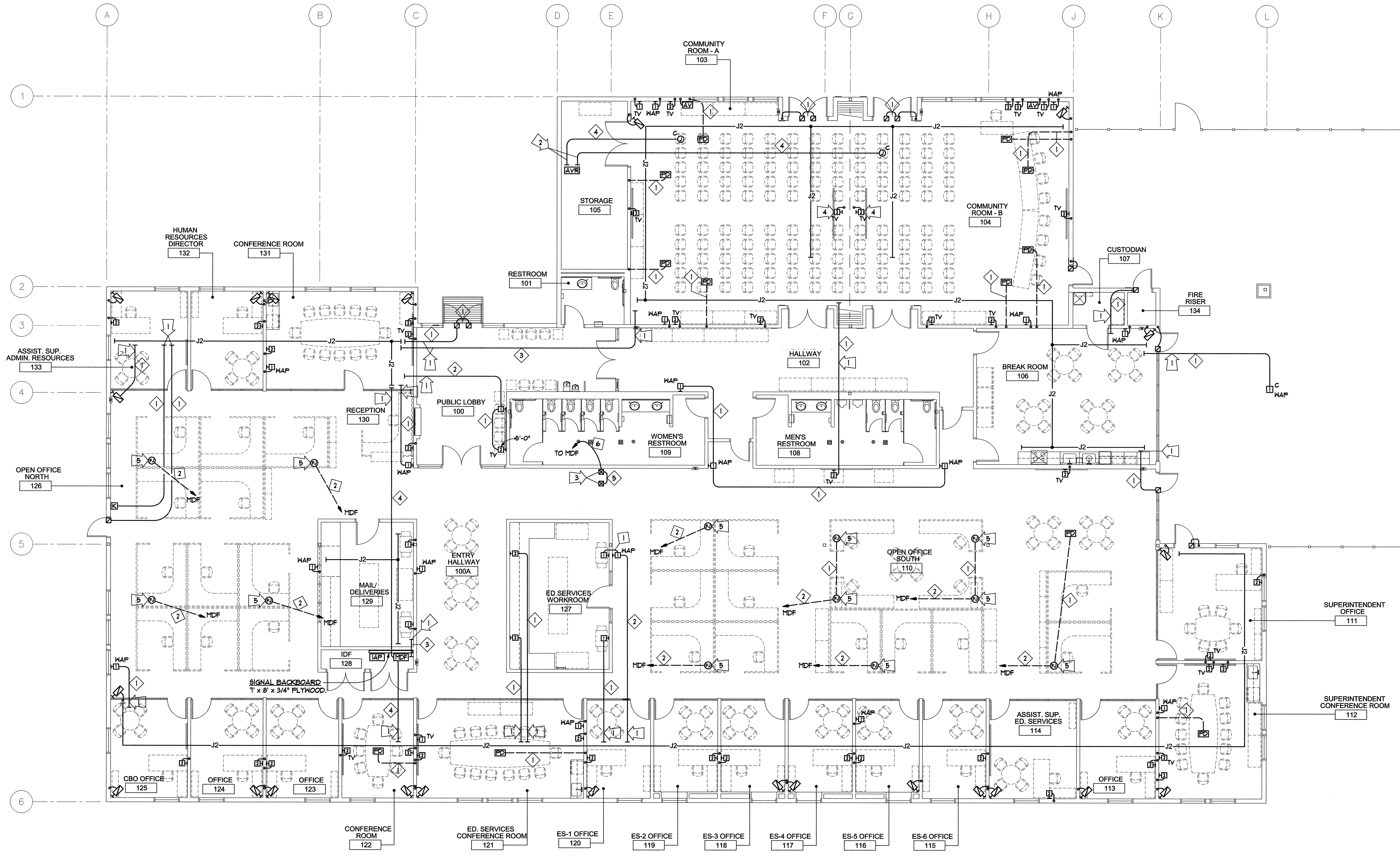
AC TM FLS HMC SS HR
DATE 2/23/17 CL

revision date: _____ by: _____

REGISTERED PROFESSIONAL ENGINEER
DAVID K. YU
No. E14887
Exp. 6/30/17
ELECTRICAL
STATE OF CALIFORNIA

600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtlewis.com
916.329.4000

ONE AND ONE-HALF INCH = ONE FOOT
ONE INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
ONE-QUARTER INCH = ONE FOOT
ONE-EIGHTH INCH = ONE FOOT
ONE-SIXTEENTH INCH = ONE FOOT
ONE INCH = TWENTY FEET
PLOT DATE: 02/22/17



SIGNAL FLOOR PLAN
E4.1
SCALE: 1/8" = 1'-0"
north

- CONSTRUCTION NOTES:**
- 1 CONDUIT STUD, SIZE AS INDICATED, ADJACENT TO THE J-HOOK CABLE SUPPORT SYSTEM.
 - 2 CONDUIT STUD AT AV RACK.
 - 3 TOP N.I.E.S. IN ATTIC SPACE.
 - 4 TV DATA OUTLET MOUNTED ON FACE OF SOFFIT.
 - 5 PROVIDE DATA JACK & CAT6 CABLES TO OFFICE FURNITURE OUTLETS. COORDINATE WITH OFFICE FURNITURE MANUFACTURER PRIOR TO BID FOR EXACT QUANTITIES.

- CONDUIT & WIRE SCHEDULE**
- 1 1-1" MTC
 - 2 1-1 1/4" MTC
 - 3 1-2" MTC
 - 4 2-2" MTC
 - 5 1"-1/2" DI
 - 6 1"-2" DI

Dry Creek Joint Elementary School District
District Education Center
8849 Cook Riolo Road, Roseville, CA 95747
Dry Creek Joint Elementary School District

FILE NO. 31-15
IDENTIFICATION STAMP DIV. OF THE
STATE ARCHITECT OFFICE OF
REGULATION SERVICES
APPLICATION NO.
02-115456
AC. TM. FLS. HYS. SS. HYP.
DATE 1/22/17 CEA

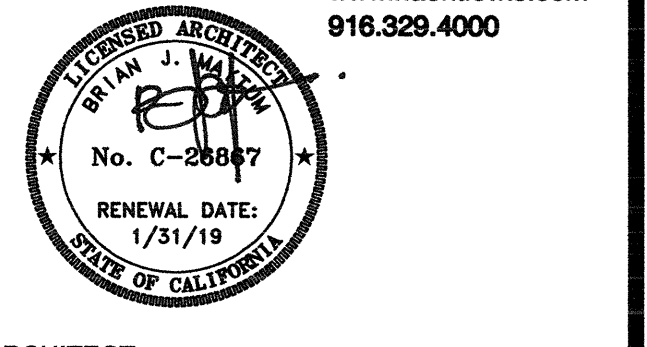
DIVISION OF THE STATE ARCHITECT
revision date: by:

HARRY A. YEE & ASSOCIATES, INC.
ELECTRICAL ENGINEERS
4920 TREETOP BLVD., SUITE D
SACRAMENTO, CALIFORNIA 95822
TEL: 916.454.5319
FAX: 916.454.4117
HVA Job #1602



CONSULTANT

nacht&lewis
600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtandlewis.com
916.329.4000



ARCHITECT

INITIAL BOX			
NO.	DWG BY	DATE	REVIEWED

REVISIONS			
NO.	DESCRIPTION	DATE	REV

DATE February 23, 2017
JOB NO. Y1538.00
SHEET TITLE

Signal Floor Plan

SHEET NO.
E4.1
SHEET ____ OF ____ TOTAL

CONSTRUCTION NOTES:
HEAT DETECTOR MOUNTED IN ATTIC MECHANICAL SPACE.

NOTES:
1. WALL MOUNTED DEVICES SHALL BE 'RED'.
2. CEILING MOUNTED DEVICES SHALL BE 'WHITE'.

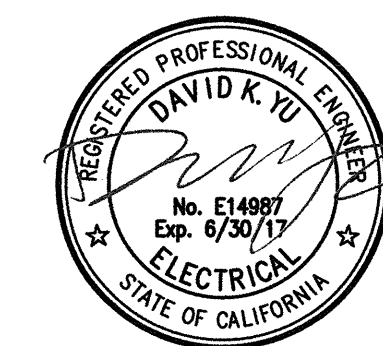
Dry Creek Joint Elementary School District
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DATE 2/23/17 C.A.

DIVISION OF THE STATE ARCHITECT
revision date: by:

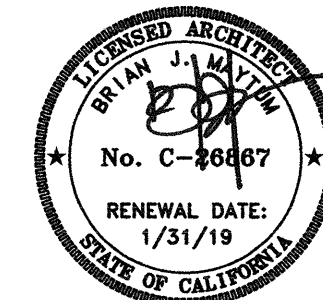
HARRY A. YEE & ASSOCIATES, INC.
ELECTRICAL ENGINEERS
4920 FREEMONT BLVD., SUITE D
SACRAMENTO, CALIFORNIA 95822
TEL: 916.454.3319
FAX: 916.454.4117
HVA Job #1602



CONSULTANT

nacht&lewis

600 Q Street, Suite 100
Sacramento, CA 95811
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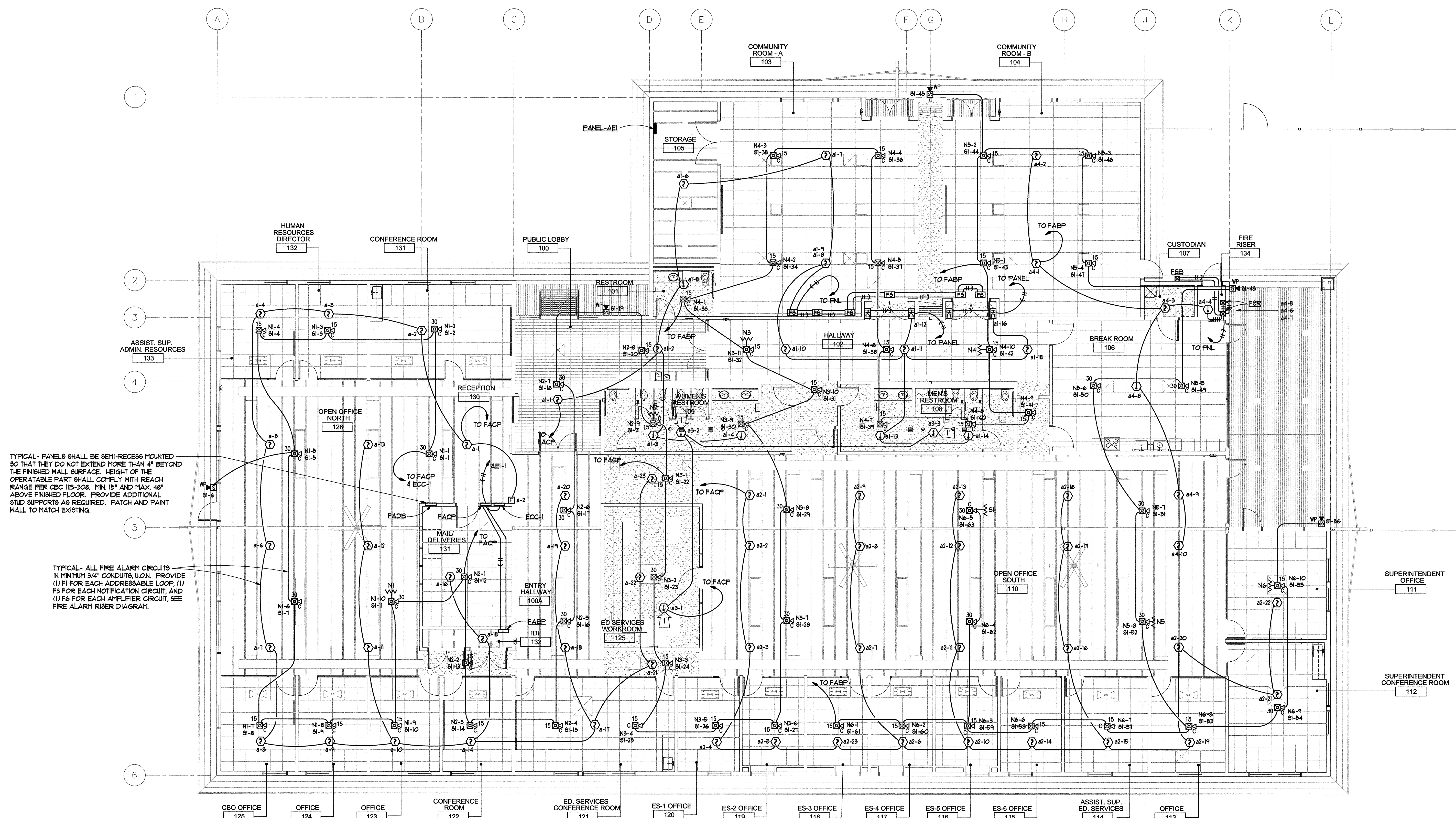
SHEET TITLE

Fire Alarm
Floor Plan

SHEET NO.

E5.1

SHEET ____ OF ____ TOTAL



TYPICAL - PANELS SHALL BE SEMI-RECESSED MOUNTED SO THAT THEY DO NOT EXTEND MORE THAN 4" BEYOND THE FINISHED WALL SURFACE. HEIGHT OF THE OPERABLE PART SHALL COMPLY WITH REACH RANGE PER CBC 11B-308, MIN. 15" AND MAX. 48" ABOVE FINISHED FLOOR. PROVIDE ADDITIONAL STUD SUPPORTS AS REQUIRED. PATCH AND PAINT WALL TO MATCH EXISTING.

TYPICAL - ALL FIRE ALARM CIRCUITS IN MINIMUM 3/4" CONDUITS, U.O.N. PROVIDE (1) F1 FOR EACH ADDRESSABLE LOOP, (1) F3 FOR EACH NOTIFICATION CIRCUIT, AND (1) F6 FOR EACH AMPLIFIER CIRCUIT, SEE FIRE ALARM RISER DIAGRAM.

A FIRE ALARM FLOOR PLAN
E5.1
SCALE: 1/8" = 1'-0"

ONE AND ONE-HALF INCH = ONE FOOT
ONE INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
ONE-QUARTER INCH = ONE FOOT
ONE-EIGHTH INCH = ONE FOOT
ONE-SIXTEENTH INCH = ONE FOOT
ONE INCH = TWENTY FEET
DATE: 02/22/17

FIRE ALARM EQUIPMENT SCHEDULE					
SYMBOL	EQUIPMENT / DEVICE	MANUFACTURER AND MODEL NUMBER	CSFM LISTING NO.	STANDBY AMPS	ALARM AMPS
	FIRE ALARM AMPLIFIER (BOM)	FIRE-LITE #ECC-30/100 W/ ECC-XRM-10V	6911-007B-0226	.372	.681
	FIRE ALARM CONTROL PANEL	EDWARDS #EST-3	7165-1651-0186	.081	.150
	FIRE ALARM BOOSTER PANEL	EST-BP66A	7300-1651-0229	.010	.190
	FIRE ALARM DOCUMENT BOX	SPACE AGE ELECTRONICS #SGC060891	.	.	.
	CONTROL RELAY MODULE	EST# 51GA-CR	7300-1651-0121	.0001	.0001
	MONITOR MODULE	EST# 51GA-CTI	7300-1651-0121	.00025	.0004
	ISOLATOR MODULE	EST# 51GA-IM	7300-1651-0121	.000045	.000045
	PULL STATION	EST# 51GA-Z10	7150-1651-0129	.000045	.000045
	DOOR HOLDER	EDWARDS #150B-AQNS	3550-1501-0131	.015	0
	SYNC. MODULE	WHEELCOCK #SM-1224-R	7300-0785-0132	0	.038
	ADDRESSABLE SMOKE DETECTOR	EST# 51GA-PS	7272-1651-0126	.000045	.000045
	ADDRESSABLE HEAT DETECTOR	EST# 51GA-HRS	7270-1651-0125	.000045	.000045
	EXTERIOR SPEAKER	EST# HG-APF-S	7300-1651-0289	0	.0286
	CEILING SPEAKER/STROBE (15cdd)	EST# GCF-SZVM	7300-1651-0211	0	.109/.00714
	CEILING SPEAKER/STROBE (30cdd)	EST# GCF-SZVM	7300-1651-0211	0	.151/.00714

FIRE ALARM WIRE SCHEDULE					
DESIG.	DESCRIPTION	USE	SYSTEM	O.D. (Inches)	AREA (sq. Inches)
F1	1 PR#16, STP. ADDRESSABLE LOOP, FFL, LON CAP, NPK #D914	BLDG., INITIATING	FIRE ALARM	0.226	0.0401
F2	1 PR#14, STP. ADDRESSABLE LOOP, FFL, MET LOC, LON CAP, NPK #A225B	SITE, INITIATING		0.35	0.0962
F3	2#12, TH4N	BLDG. SIGNAL		0.13 EA.	0.0265
F4	2#10, TH4N	SITE, SIGNAL		0.164 EA.	0.0422
F5	2#14 FFL	HARDWIRED INT.			
F6	1 PR#16 STRAND, FFLR MID-CAP, NPK #9106	BLDG., SPEAKER		0.182 EA.	
F7	1 PR#16 STRAND, FFL, MET LOC, NPK #A225B	SITE, SPEAKER		0.295 EA.	
F8	6 CONDUCTOR CABLE	REMOTE PAGER UNIT			

BATTERY CALCULATION WORKSHEET					
EQUIPMENT / DEVICE	QUANTITY	DEVICE STANDBY CURRENT (AMPS)	TOTAL STANDBY CURRENT (AMPS)	DEVICE ALARM CURRENT (AMPS)	TOTAL ALARM CURRENT (AMP)
FIRE ALARM BOOSTER PANEL-FABP	1	.010	.010	.190	.190
FIRE ALARM CEILING SPEAKER/STROBE (15cdd)	21	0	0	.109	2.289
FIRE ALARM CEILING SPEAKER/STROBE (30cdd)	7	0	0	.151	1.057
FIRE ALARM SYNC. MODULE	3	0	0	.038	.114
TOTALS:		.010 AMPS		3.65 AMPS	

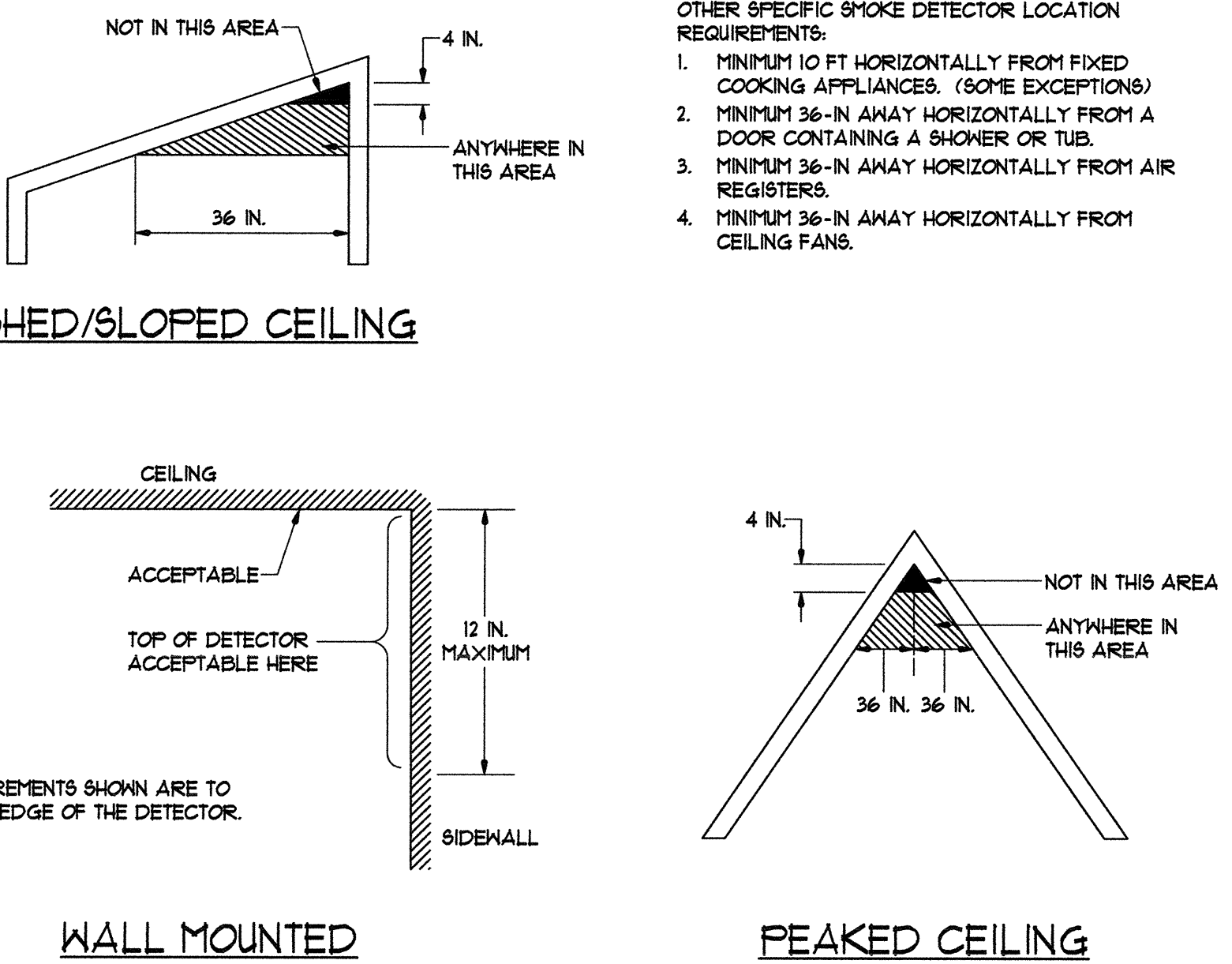
NOTES:
BATTERY CALCULATIONS FOR 24 HOURS (STANDBY) PLUS 15 MINUTES (ALARM):
(TOTAL STANDBY CURRENT) (24 HRS) + (TOTAL ALARM CURRENT) ($\frac{15}{60}$ H) x (25%) = MIN. BATTERY AMP HRS. REQ'D.
(.01) (24 HRS) + (3.65) ($\frac{15}{60}$ H) x (25%) = 3.24 AMP HOUR (MINIMUM BATTERY)
BATTERY PROVIDED = 1.0 AH.

BATTERY CALCULATION WORKSHEET					
EQUIPMENT / DEVICE	QUANTITY	DEVICE STANDBY CURRENT (AMPS)	TOTAL STANDBY CURRENT (AMPS)	DEVICE ALARM CURRENT (AMPS)	TOTAL ALARM CURRENT (AMP)
FIRE ALARM AMPLIFIER PANEL-ECC	1	.372	.372	.681	.681
FIRE ALARM EXTERIOR SPEAKER	5	0	0	.0286	.143
FIRE ALARM CEILING SPEAKER/STROBE (15cdd)	39	0	0	.00714	.2785
FIRE ALARM CEILING SPEAKER/STROBE (30cdd)	19	0	0	.00714	.136
TOTALS:		.372 AMPS		1.24 AMPS	

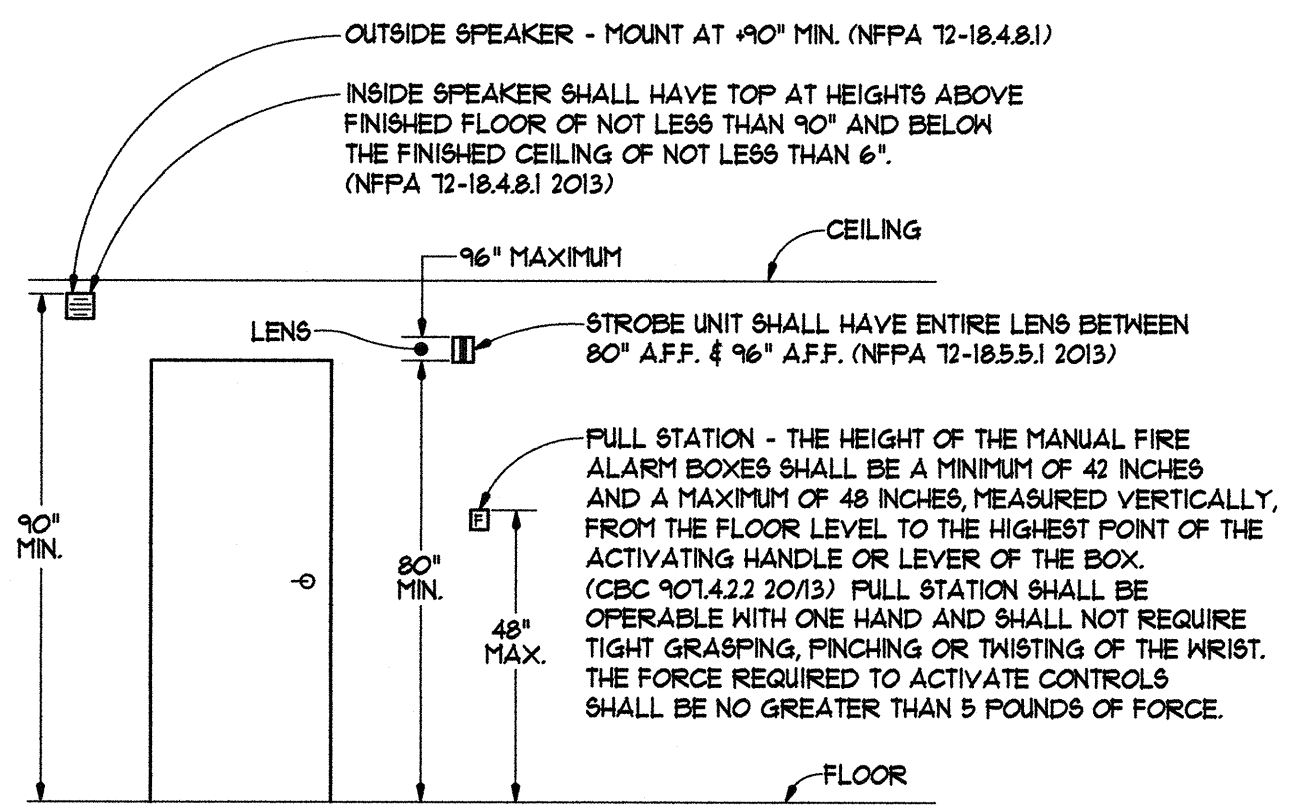
NOTES:
BATTERY CALCULATIONS FOR 24 HOURS (STANDBY) PLUS 15 MINUTES (ALARM):
(TOTAL STANDBY CURRENT) (24 HRS) + (TOTAL ALARM CURRENT) ($\frac{15}{60}$ H) x (25%) = MIN. BATTERY AMP HRS. REQ'D.
(.372) (24 HRS) + (1.24) ($\frac{15}{60}$ H) x (25%) = 11.55 AMP HOUR (MINIMUM BATTERY)
BATTERY PROVIDED = 18.0 AH.

BATTERY CALCULATION WORKSHEET					
EQUIPMENT / DEVICE	QUANTITY	DEVICE STANDBY CURRENT (AMPS)	TOTAL STANDBY CURRENT (AMPS)	DEVICE ALARM CURRENT (AMPS)	TOTAL ALARM CURRENT (AMP)
FIRE ALARM CONTROL PANEL-FACP	1	.081	.081	.150	.150
FIRE ALARM SMOKE DETECTOR	59	.000045	.002655	.000045	.002655
FIRE ALARM HEAT DETECTOR	10	.000045	.00045	.000045	.00045
PULL STATION	1	.00025	.00025	.0004	.0004
MONITOR MODULE	5	.00025	.00125	.0004	.0020
CONTROL MODULE	6	.0001	.0006	.0001	.0006
ISOLATOR MODULE	10	.000045	.00045	.000045	.00045
DOOR HOLDER	4	.015	.06	0	0
FIRE ALARM CEILING SPEAKER/STROBE (15cdd)	18	0	0	.109	1.962
FIRE ALARM CEILING SPEAKER/STROBE (30cdd)	12	0	0	.151	1.812
FIRE ALARM SYNC. MODULE	3	0	0	.038	.114
TOTALS:			.141 AMPS		4.045 AMPS

NOTES:
BATTERY CALCULATIONS FOR 24 HOURS (STANDBY) PLUS 15 MINUTES (ALARM):
(TOTAL STANDBY CURRENT) (24 HRS) + (TOTAL ALARM CURRENT) ($\frac{15}{60}$ H) x (25%) = MIN. BATTERY AMP HRS. REQ'D.
(.141) (24 HRS) + (4.045) ($\frac{15}{60}$ H) x (25%) = 5.67 AMP HOUR (MINIMUM BATTERY)
BATTERY PROVIDED = 1.0 AH.



SMOKE DETECTOR LOCATION REQ'S
NO SCALE



PULL STATION, STROBE & SPEAKER UNIT ELEVATION DETAIL
NO SCALE

FIRE ALARM SYSTEM NOTES:

- THE FIRE ALARM SYSTEM IS A COMPLETE PLAN SUBMITTAL.
- THE FIRE ALARM SYSTEM SHALL CONFORM TO THE CALIFORNIA BUILDING CODE (CBC) SECTION 901, THE CALIFORNIA ELECTRICAL CODE (CEC) ARTICLE 160 AND THE CALIFORNIA FIRE CODE (CFC) SECTION 901.
- UPON COMPLETION OF THE INSTALLATION OF THE FIRE ALARM SYSTEM A SATISFACTORY TEST OF THE ENTIRE SYSTEM SHALL BE MADE IN THE PRESENCE OF THE DIVISION OF THE STATE ARCHITECT (DSA) INSPECTOR OF RECORD.
- INSTALLATION OF THE FIRE ALARM SYSTEM SHALL NOT BE STARTED UNTIL DETAILED PLANS AND SPECIFICATIONS INCLUDING CALIFORNIA STATE FIRE MARSHAL LISTING NUMBERS FOR EACH COMPONENT OF THE SYSTEM, VOLTAGE DROP CALCULATIONS, AND BATTERY CALCULATIONS HAVE BEEN SUBMITTED AND APPROVED. IF THE INFORMATION ON THE BID DRAWINGS ARE REUSED AS THE SUBMITTAL DRAWINGS, GOOD ALL CHANGES.
- ALL CEILING HEAT AND SMOKE DETECTORS SHALL BE LOCATED AWAY FROM THE SUPPLY AIR REGISTER, DISTANCE AS RECOMMENDED BY EQUIPMENT MANUFACTURER.
- BASIC SOUND AND PATTERN OF THE NEW FIRE ALARM AUDIBLE DEVICES SHALL SOUND THE CALIFORNIA FIRE ALARM SIGNAL IN TEMPORAL MODE.
- PROVIDE COMPLETE SYSTEM WIRING. ALL WIRING SHALL BE IN MINIMUM 3/4" CONDUIT. NO OTHER SYSTEMS WIRING SHALL BE ROUTED IN THE FIRE ALARM CIRCUITS. HARDWIRED INITIATING CIRCUITS SHALL BE MINIMUM #14 THAW/THAN. SYSTEMS WIRING SHALL BE PER EQUIPMENT MANUFACTURER'S RECOMMENDATIONS AND THEIR REQUIREMENTS.
- THE FIRE ALARM SYSTEM SHALL COMPLY WITH A TOTAL COVERAGE FULLY AUTOMATIC SYSTEM. DETECTORS SHALL BE INSTALLED IN ALL AREAS WHERE REQUIRED BY THE APPROPRIATE NFPA STANDARD OR THE AUTHORITY HAVING JURISDICTION. EACH INSTALLED DETECTOR SHALL BE ACCESSIBLE FOR PERIODIC MAINTENANCE AND TESTING. PROVIDE ACCESS DOORS AS REQUIRED. TOTAL COVERAGE SHALL INCLUDE ALL ROOMS, STORAGE AREAS, ATTICS, SPACES ABOVE SUSPENDED CEILINGS, AND OTHER SUBDIVISIONS AND ACCESSIBLE SPACES. INACCESSIBLE AREAS SHALL NOT BE REQUIRED TO BE PROTECTED BY DETECTORS UNLESS THEY CONTAIN COMBUSTIBLE MATERIAL, IN WHICH CASE THEY SHALL BE MADE ACCESSIBLE AND BE PROTECTED BY DETECTOR(S).
- ALARM-INDICATING DEVICES OF A FIRE ALARM SYSTEM INTENDED TO ALERT ALL OCCUPANTS SHALL CAUSE A LEVEL OF AUDIBILITY OF NOT LESS THAN 15 DBA ABOVE THE AVERAGE AMBIENT NOISE LEVELS OR 5 DBA ABOVE THE MAXIMUM SOUND LEVEL HAVING A DURATION OF 60 SECONDS WHICH-EVER IS GREATER MEASURED 3'-0" ABOVE THE FLOOR. AMBIENT NOISE LEVELS MEANS THE LEVEL WHICH CAN NORMALLY BE EXPECTED WHEN THE FACILITY, BUILDING, ROOM OR AREA IS FUNCTIONING UNDER NORMAL OPERATING OR WORKING CONDITIONS (NFPA 72, SEC. 18.4.3.1).
- THE AUTOMATIC FIRE ALARM SYSTEM SHALL COMPLY WITH SENATE BILL NO. 575 (SB575) & CEC 901.2.3. FOR NEW CONSTRUCTION PROJECTS.
- THE LOCATION OF AUTOMATIC DETECTORS, MANUAL STATIONS AND OTHER FIRE ALARM EQUIPMENT AND DEVICES, AS SHOWN ON PLAN, ARE FOR REFERENCE ONLY AND DO NOT CONSTITUTE SHOP DRAWINGS WHICH ARE REQUIRED FOR REVIEW AND APPROVAL.
- THE ALARM SYSTEM SHALL ACTIVATE A MEANS OF WARNING THE HEARING IMPAIRED. FLASHING VISUAL WARNINGS SHALL HAVE A FLASH RATE NOT EXCEEDING TWO FLASHES PER SECOND (2 HZ) NOR BE LESS THAN ONE FLASH EVERY SECOND (1 HZ). STROBE SIGNALING DEVICES FOR THE HEARING IMPAIRED SHALL BE STATE FIRE MARSHAL APPROVED AND LISTED (NFPA 72, SEC. 18.5.2.1).
- POWER SERVICE SHALL BE ON A DEDICATED BRANCH CIRCUIT WITH A RED MARKING AND IDENTIFIED AS "FIRE ALARM CIRCUIT CONTROL".
- AUDIBLE FIRE ALARM SOUND LEVEL SHALL BE AT LEAST 15 DBA ABOVE AVERAGE AMBIENT SOUND LEVEL IN ALL OCCUPABLE AREAS. (16 CLASSROOM AVERAGE AMBIENT ROOM NOISE IS 45 DBA PLUS 15 DBA EQUALS 60 DBA MINIMUM ALARM TONE REQUIRED). SYNCHRONIZE AUDIBLE SIGNAL PER NFPA 72, 18.4.1.2.
- STROBES SHALL FLASH AT A RATE NOT EXCEEDING TWO FLASHES PER SECOND NOR BE LESS THAN ONE FLASH EVERY SECOND. SYNCHRONIZE STROBE FLASH PER NFPA 72, 18.5.3.4.2.
- CBC Sec. 901.5.2: AUDIBLE SIGNALS INTENDED FOR OPERATION IN THE PUBLIC MODE SHOULD HAVE A SOUND LEVEL OF NOT LESS THAN 15 DBA AT 10 FEET OR MORE THAN 110 DBA AT THE MINIMUM HEARING DISTANCE FROM THE AUDIBLE APPLIANCE.
- FINAL FIRE ALARM TEST SHALL BE MADE WITH THE DSA INSPECTOR OF RECORD (IOR). LOCAL FIRE AUTHORITY SHALL BE ADVISED DATE AND TIME OF FINAL FIRE ALARM TESTING AND SHALL ASSISTANCE SUCH TESTING WHEN ABLE.
- THE INSTALLING CONTRACTOR SHALL PROVIDE A RECORD OF COMPLETION PER NFPA 72, SECTION 14.6, FIGURE 14.6.2.4 & CEC 901.12 TO THE INSPECTOR OF RECORD (IOR) / DSA AFTER COMPLETION OF OPERATIONAL ACCEPTANCE TESTS.
- WHEN THE SPECIFIED EQUIPMENT IS SUBSTITUTED OR ANY CHANGES ARE MADE THE CONTRACTOR SHALL OBTAIN DSA APPROVAL VIA THE DSA CHANGE ORDER PROCESS PER DSA IR A-21.

FIRE ALARM RECORD DOCUMENTS CABINET NFPA 72, 7.1.2

- EVERY NEW FIRE ALARM SYSTEM SHALL PROVIDE A DOCUMENTATION CABINET, INSTALLED AT THE SYSTEM CONTROL, PANEL OR OTHER APPROVED LOCATION.
- THE DOCUMENTATION CABINET SHALL BE PROMINENTLY LABELED, "SYSTEM RECORD DOCUMENTS".
- ALL RECORD AND TESTING DOCUMENTATION SHALL BE STORED IN THE CABINET.
- CONTENTS SHALL BE ACCESSIBLE BY AUTHORIZED PERSONNEL ONLY.
- WHERE CABINET IS INSTALLED IN A LOCATION OTHER THAN THE SYSTEM CONTROL UNIT, ITS LOCATION SHALL BE IDENTIFIED AT THE SYSTEM CONTROL UNIT.
- SYSTEM DOCUMENTS AS APPLICABLE:
 - RECORD DRAWINGS / AS-BUILTS
 - EQUIPMENT OUT SHEETS & CA 981 LISTINGS
 - ALTERNATIVE MEANS AND METHODS
 - PERFORMANCE BASED DESIGN DOCUMENTATION (NFPA 72, 13.1)
 - SYSTEM RECORD OF COMPLETION & ANY SUPPLEMENTAL INSPECTION AND TESTING DOCUMENTATION (NFPA 72, 18.2)
 - EMERGENCY RESPONSE PLAN (NFPA 72, 13.8)
 - EVALUATION DOCUMENTATION (NFPA 72, 13.4)
 - RISK ANALYSIS DOCUMENTATION (NFPA 72, 13.6)
 - SOFTWARE & FIREWARE CONTROL DOCUMENTATION (NFPA 72, 23.22)

FIRE ALARM DEVICE SEQUENCE OF OPERATION MATRIX

	INITIATION	MANUAL PULL STATION	AREA SMOKE DETECTORS	HEAT DETECTOR	DUCT DETECTOR	ELEVATOR RECALL	TROUBLE CONDITION (SHORT, OPEN, POWER FAILURE)
ANNUNCIATE AT FACP	YES	YES	YES	YES	YES	YES	YES
ACTIVATE AUDIBLE & VISIBLE ALARMS THROUGH OUT CAMPUS	YES	YES	YES	YES	YES	YES	NO
ACTIVATE EVAC SYSTEM THROUGH OUT CAMPUS	YES	YES	YES	YES	YES	YES	NO

NOTES:

- AFFICABLE STANDARD 2013 NFPA 72.
- A STAMPED SET OF APPROVED FIRE ALARM DESIGN DOCUMENTS SHALL BE ON THE JOB SITE AND USED FOR INSTALLATION.
- ANY DISCREPANCIES BETWEEN THE DRAWINGS AND THE CODE OR RECOGNIZED STANDARDS SHALL BE BROUGHT TO THE ATTENTION OF DSA AND THE ARCHITECT/ENGINEER OF THE PROJECT.
- DSA ARCHITECT/ENGINEER AND OWNER SHALL BE NOTIFIED A MINIMUM OF 48 HOURS PRIOR TO THE FINAL INSPECTION AND/OR TESTING.
- ALL PENETRATIONS THROUGH RATED ASSEMBLIES, REQUIRING OPENING PROTECTION SHALL BE PROVIDED WITH A PENETRATION FIRE STOP SYSTEM AS IDENTIFIED IN CBC CHAPTER 7.15, OR OTHER LAB TESTING CRITERIA. APPROVED TYPE OF MATERIALS SHALL BE IDENTIFIED WITHIN THE SPECIFICATION WITHIN THE FIRE ALARM SECTION.
- WALL MOUNTED VISUAL NOTIFICATION DEVICES SHALL HAVE THEIR ENTIRE LENS MOUNTED WITHIN 80" MINIMUM AND 96" MAXIMUM FROM FINISHED FLOOR.
- WALL MOUNTED AUDIBLE NOTIFICATION DEVICES SHALL HAVE THEIR TOPS MOUNTED AT 90" MINIMUM AND 100" MAXIMUM FROM FINISHED FLOOR AND NO CLOSER THEN 6" TO A HORIZONTAL STRUCTURE.
- AUDIBLE DEVICES TO BE AT LEAST 15 DBA ABOVE THE AVERAGE AMBIENT SOUND LEVEL BUT NOT LESS THAN 15 DBA AT 10 FEET OR MORE THAN 110 DBA AT THE MINIMUM HEARING DISTANCE SOUND LEVEL SHALL BE MAINTAINED FOR DURATION OF AT LEAST 60 SECONDS 5 DBA MUST BE MAINTAINED.
- THE CONTRACTOR SHALL ADJUST/INSTALL ALL DEVICES TO MAXIMIZE PERFORMANCE AND TO MINIMIZE FALSE ALARMS.
- UNDERGROUND AND EXTERIOR CONDUITS TO HAVE WATERTIGHT FITTINGS AND WIRE TO BE APPROVAL FOR NET LOCATIONS.
- ALL FIRE ALARM WIRING SHALL BE FLUP OR FLPL (FIRE POWER LIMITED OR FIRE POWER LIMITED PLENFI) AS REQUIRED FOR APPLICATION. WIRING IN CONDUIT ABOVE GROUND MAY BE THIN OR THIN.
- PER CEC STANDARDS. ALL WIRING IS TO BE PULLED THROUGH EACH JUNCTION BOX AND CONNECTED DIRECTLY TO EACH FIRE DEVICE. DO NOT SPICE THE WIRE. THERE MUST BE AT LEAST 6' OF LEAD WIRE FROM THE BOX TO THE DEVICE. ALL BOXES TO BE SIZED PER CEC.
- ALL FIRE ALARM CIRCUITS SHALL BE IN CONDUIT, SURFACE RACEWAY OR OPEN RUN ABOVE CEILINGS, UNDER FLOORS AND IN WALLS IN A NEAT AND PROTECTED MANNER AS INDICATED ON DESIGN DOCUMENTS. EXPOSED CIRCUITS ARE ONLY PERMITTED WHEN NOTED AS EXPOSED ON DESIGN DOCUMENTS.
- FIRE ALARM PANEL SHALL BE SECURED TO MOUNTING SURFACES PER MANUFACTURERS SPECIFICATION. NO SINGLE DEVICE SHALL EXCEED THE HEIGHT OF 20 LBS. WITHOUT SPECIAL MOUNTING DETAILS.
- A DEDICATED BRANCH CIRCUIT SHALL BE PROVIDED FOR FIRE ALARM EQUIPMENT. THIS CIRCUIT SHALL BE ENERGIZED FROM THE COMMON USE AREA PANEL AND SHALL HAVE NO OTHER OUTLETS. THE BREAKER SHALL BE RED WITH LOCKING DEVICE TO BLOCK THE HANDLE IN THE "ON" POSITION. THE CIRCUIT BREAKER SHALL BE LABELED "FIRE ALARM CIRCUIT CONTROL". CIRCUIT ID TO BE LABELED AT FIRE PANEL/EXTENDER.
- CONTROL PANELS, REMOTE ANNUNCIATORS SHALL BE INSTALLED WITH THEIR BOTTOMS MOUNTED AT 48".
- THE INSTALLING CONTRACTOR SHALL PROVIDE SYSTEM PROGRAMMING FOR SUPERVISORY MONITORING PER CBC SECTION 901.6.2.
- SUPERVISORY MONITORING SHALL BE TESTED AND VERIFIED AS SENDING CORRECT SIGNALS IN CONJUNCTION WITH FINAL ACCEPTANCE TEST.
- OWNER SHALL BE RESPONSIBLE FOR ESTABLISHING A FIRE SYSTEM MONITORING CONTRACTOR PROVISIONS.
- HIRING AND MATERIALS PER CEC 160.

FIRE ALARM SYSTEM DESCRIPTION

THE SYSTEM IS POWER LIMITED, AUTOMATIC, MANUAL, LOCAL, AUXILIARY, REMOTE STATION (PROTECTED PREMISES) PROPRIETARY (PROTECTED PREMISES), WATERLOON, AND SPRINKLER SUPERVISORY SERVICE. SUITABLE FOR RELEASING DEVICE SERVICE, REFER TO LISTED DATA SHEET FOR DETAILED PRODUCT DESCRIPTION AND OPERATIONAL CONSIDERATIONS.

FIRE ALARM PROJECT NOTES:

- BEFORE REQUESTING FINAL APPROVAL OF THE INSTALLATION THE INSTALLING CONTRACTOR SHALL FURNISH A WRITTEN STATEMENT TO THE PROJECT INSPECTOR TO THE EFFECT THAT THE SYSTEM HAS BEEN INSTALLED AND TESTED IN ACCORDANCE WITH THE (2013) NFPA 72 SECTION 14.4.1.
- AUTOMATIC FIRE ALARM SYSTEMS SHALL TRANSMIT THE ALARM SUPERVISORY AND TROUBLE SIGNALS TO AN APPROVED SUPERVISING STATION AS REQUIRED BY NFPA 72 AND CEC 901.6.3. THE SUPERVISING STATION SHALL BE LISTED AS EITHER ULFV OR ULIS BY UL OR SHALL MEET THE REQUIREMENTS OF FM STANDARD 803.
- THE FIRE ALARM CONTRACTOR SHALL PROVIDE A SUFFICIENT NUMBER OF SPEAKERS AND QUANTITY OF STROBES AND LOCATIONS TO MEET THE AUDIBILITY AND VISUAL REQUIREMENTS OF 2013 NFPA 72. SPEAKERS AND STROBES SHOWN ON THE DRAWINGS ARE A MINIMUM AND SHALL NOT BE REDUCED FROM THE FINAL INSTALLATION.
- SPEECH INTELLIBILITY TESTING SHALL BE PERFORMED PER THE IEC 60849 STANDARD USING THE OBJECTIVE ASSESSMENT BASED ON PHYSICAL PARAMETERS OF THE TRANSMISSION CHANNEL UTILIZING SPECIALIZED METERING EQUIPMENT, THE SPEECH TRANSMISSION INDEX (STI) AND COMMON INTELLIBILITY SCALE (CIS) METHODS SHALL BE UTILIZED. THE AREAS OF HIGH INTELLIBILITY SHALL BE SCORED ABOVE .8 STI AND .9 CIS. ALL OTHER AREAS SHALL SCORE ABOVE .55 STI AND .74 CIS.
- AFTER INSTALLATION AND TESTING OF THE FIRE ALARM SYSTEM AN INDEPENDENT AUDIBILITY TEST SHALL BE PERFORMED BY A QUALIFIED THIRD PARTY TESTING AGENCY PROVIDED BY THE CONTRACTOR PRIOR TO ACCEPTANCE OF THE SYSTEM. ANY CHANGES REQUIRED DUE TO THE RESULTS OF THE AUDIBILITY TESTING SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR TO CORRECT AS NECESSARY TO BRING THE SYSTEM INTO 2013 NFPA 72, ANNEX D COMPLIANCE.
- ALL FIRE ALARM CABLE SHALL BE INSTALLED IN CONDUIT. ALL CONDUIT SHALL BE INSTALLED WITH COMPRESSION FITTINGS FOR COUPLERS AND BOX CONNECTORS.
- THE FIRE ALARM SYSTEM IS AN ADDRESSABLE SYSTEM.

Dry Creek Joint Elementary School District
District Education Center
8849 Cook Riolo Road, Roseville, CA 95747
Dry Creek Joint Elementary School District

FILE NO. 31-15
IDENTIFICATION STAMP OF THE STATE ARCHITECT OFFICE OF REGULATION SERVICES
APPLICATION NO. 02-115456
AC. TM. PLS. HMCSS. RRP
DATE 2/23/17 C.AZ

DIVISION OF THE STATE ARCHITECT
revision date: by:

HARRY A. YEE & ASSOCIATES, INC.
ELECTRICAL ENGINEERS
4520 FREEMONT BLVD., SUITE 0
SACRAMENTO, CALIFORNIA 95822
TEL: 916.454.5319
FAX: 916.454.4117
HYA Job #1602

REGISTERED PROFESSIONAL ENGINEER
DAVID A. YEE
No. E14897
Exp. 6/30/17
ELECTRICIAN
STATE OF CALIFORNIA
CONSULTANT

nacht&lewis
800 Q Street, Suite 100
Sacramento, CA 95811
www.nachtlewis.com
916.329.4000

REGISTERED ARCHITECT
DAVID A. YEE
No. C-20867
RENEWAL DATE: 1/31/19
ARCHITECT

INITIAL BOX			
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REVISIONS			
NO.	DESCRIPTION	DATE	REV'D
	100% design development	7/6/16	
	75% construction documents	9/20/16	
	DSA submittal	12/31/16	
	DSA approval	02/23/17	

DATE February 23, 2017

JOB NO. Y1538.00

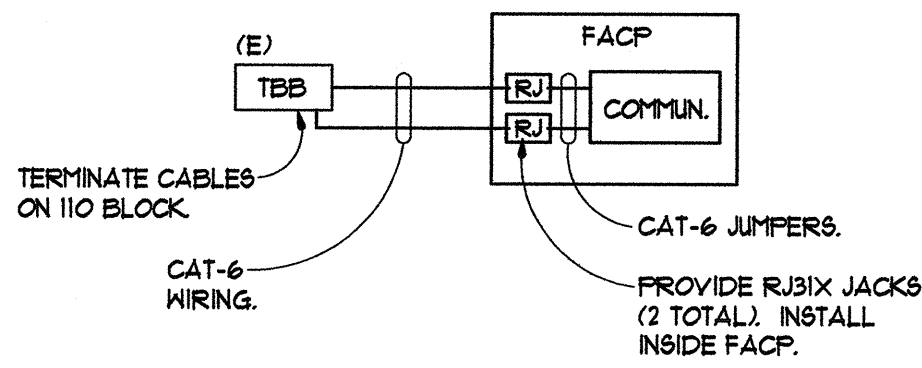
SHEET TITLE
Fire Alarm
Schedules,
Calculations,
Notes & Details

SHEET NO.

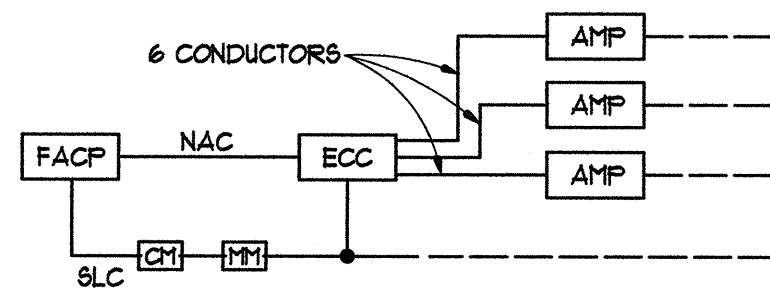
E5.2

SHEET OF TOTAL

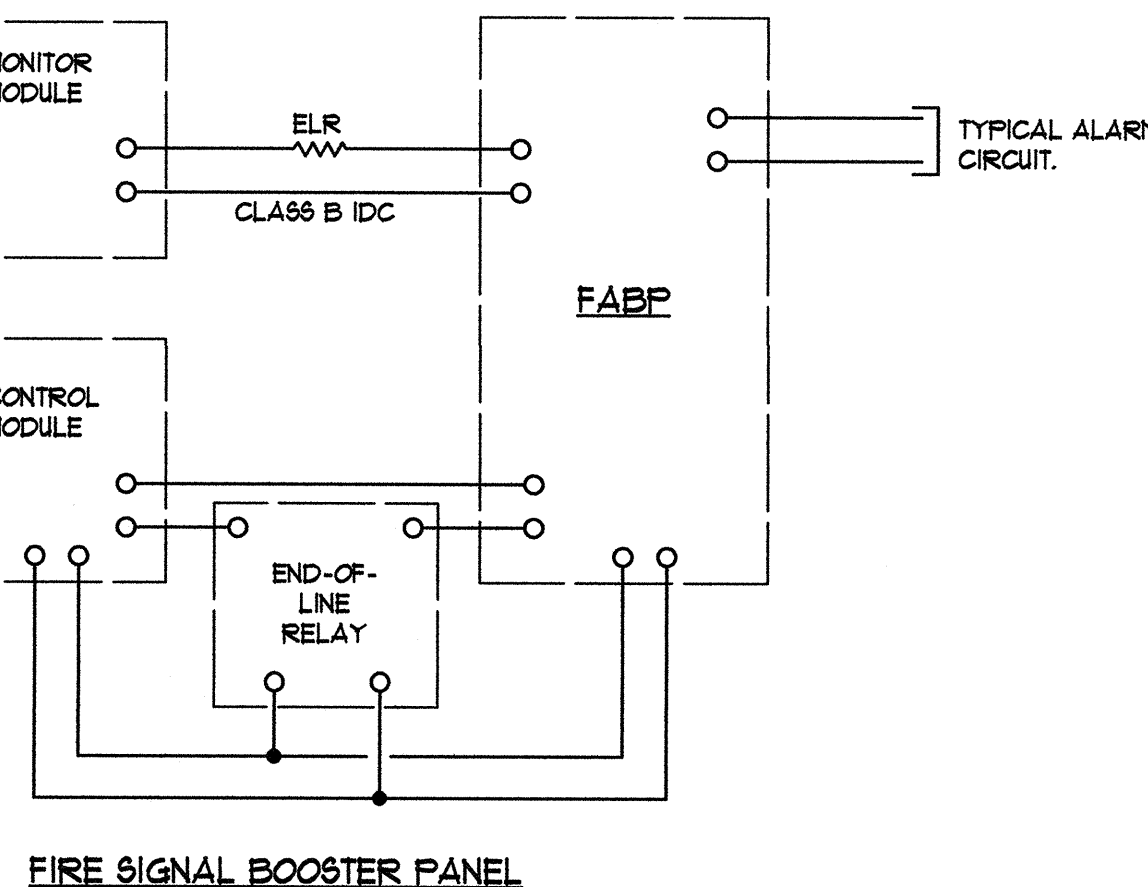
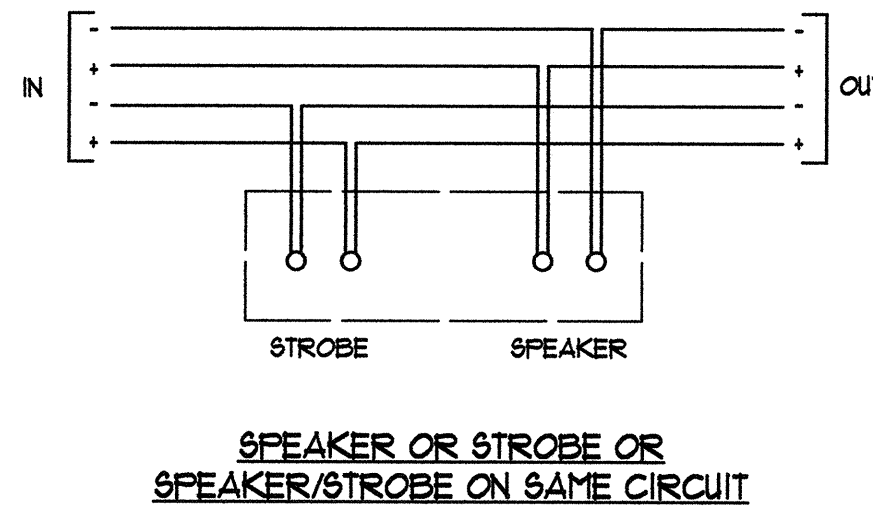
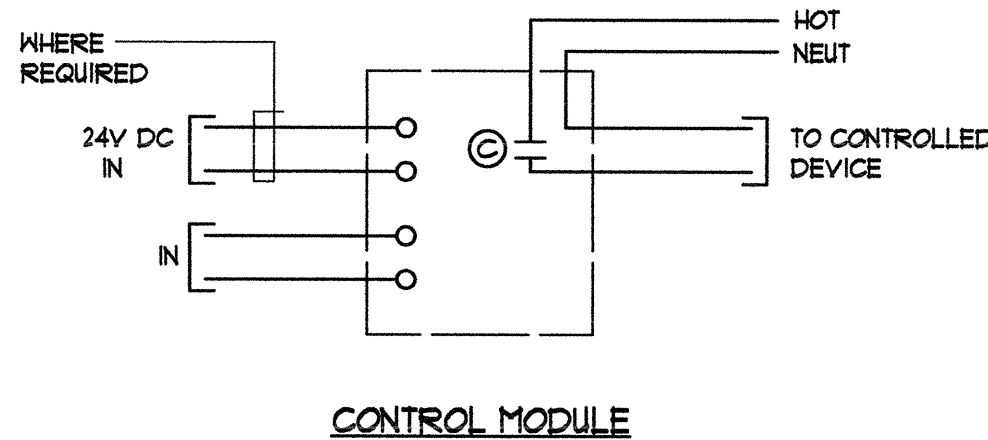
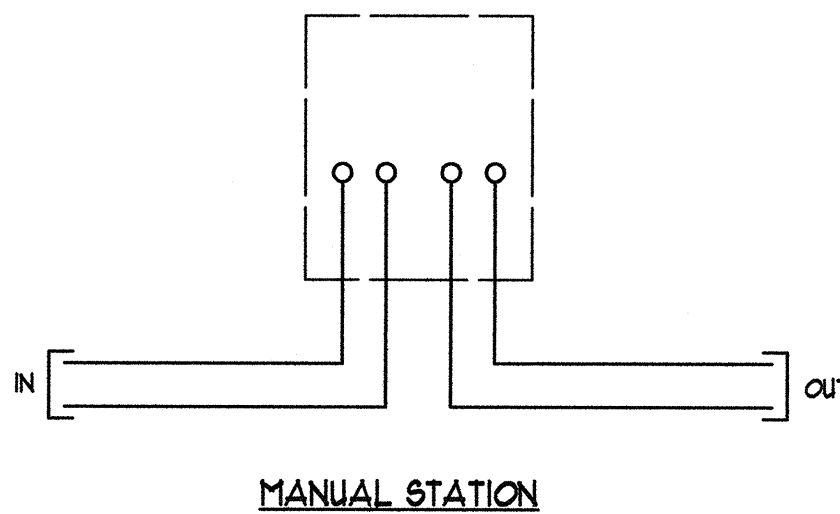
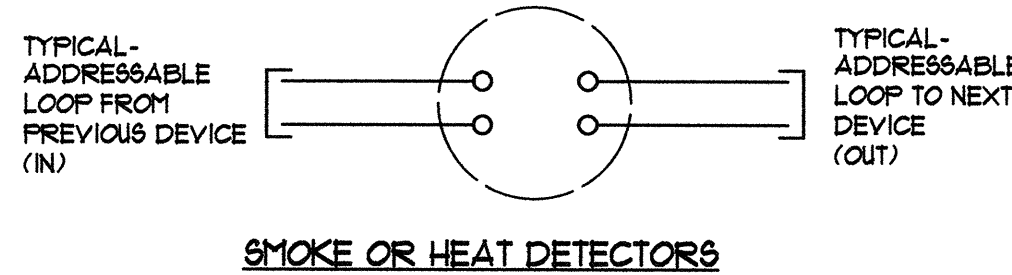
PLT DATE: 02/22/17
ONE INCH = TWENTY FEET
ONE-SIXTEENTH INCH = ONE FOOT
ONE-EIGHTH INCH = ONE FOOT
ONE-FOURTH INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE INCH = ONE FOOT
ONE AND ONE-HALF INCH = ONE FOOT



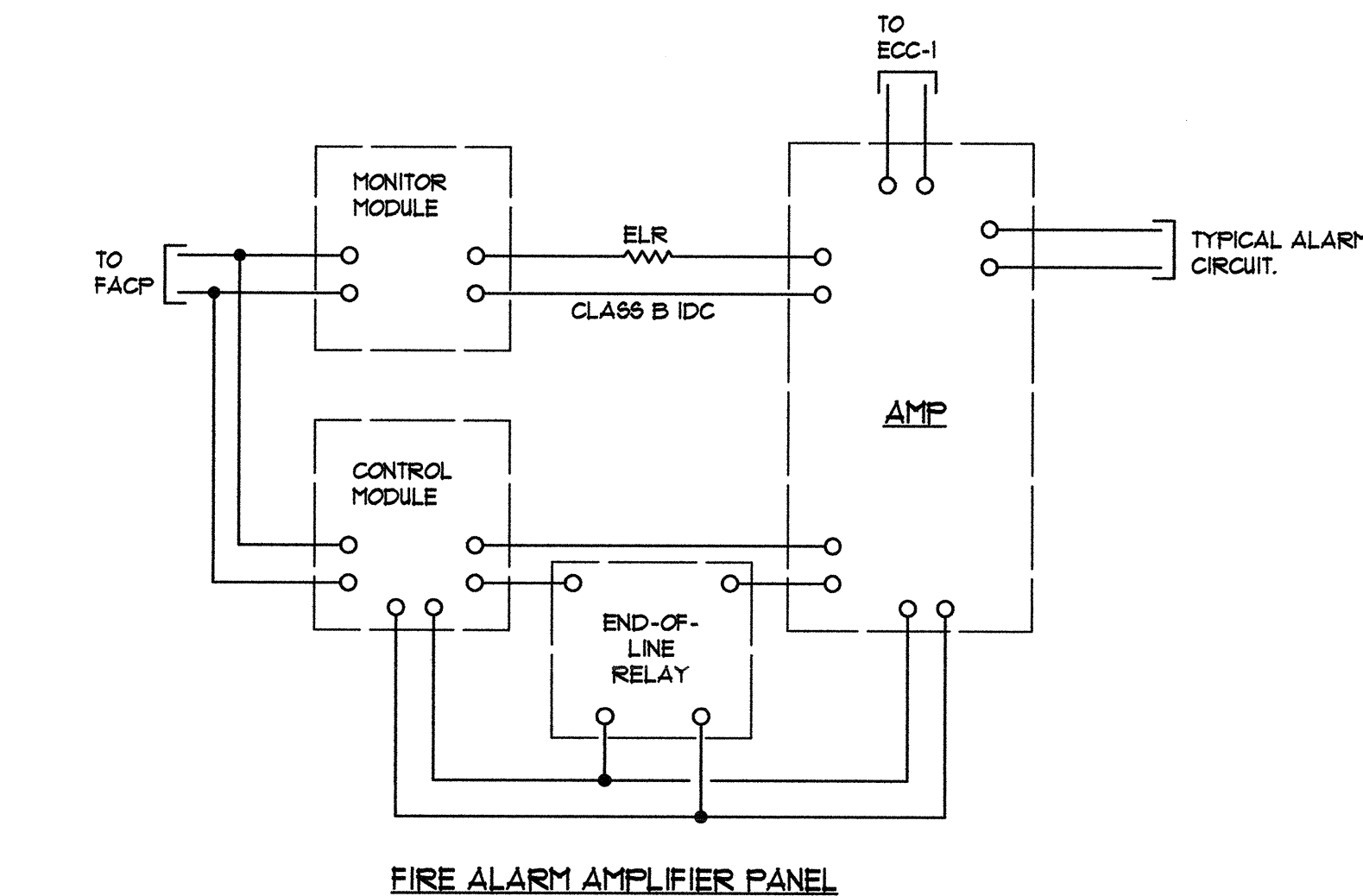
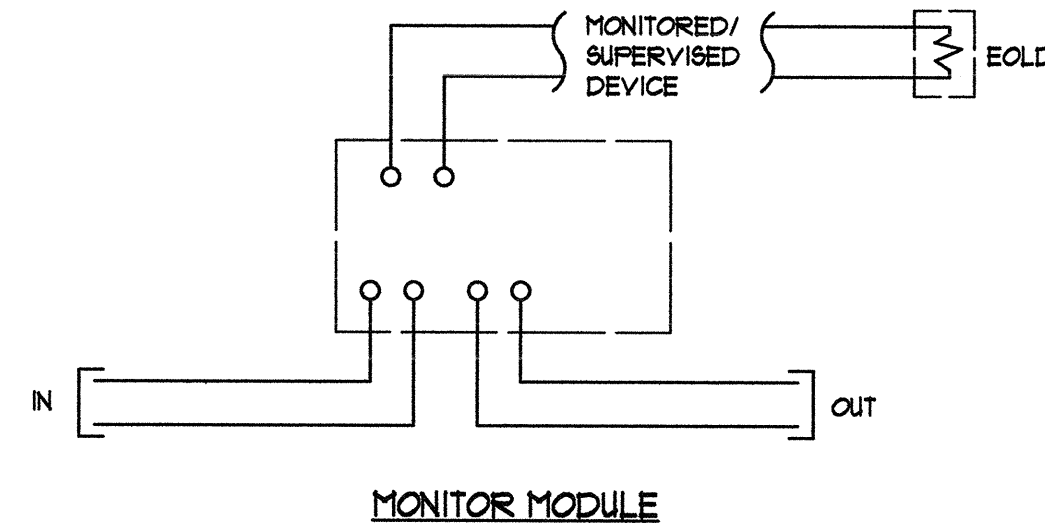
C
E5.4
FACP TELEPHONE WIRING DIAGRAM
NO SCALE



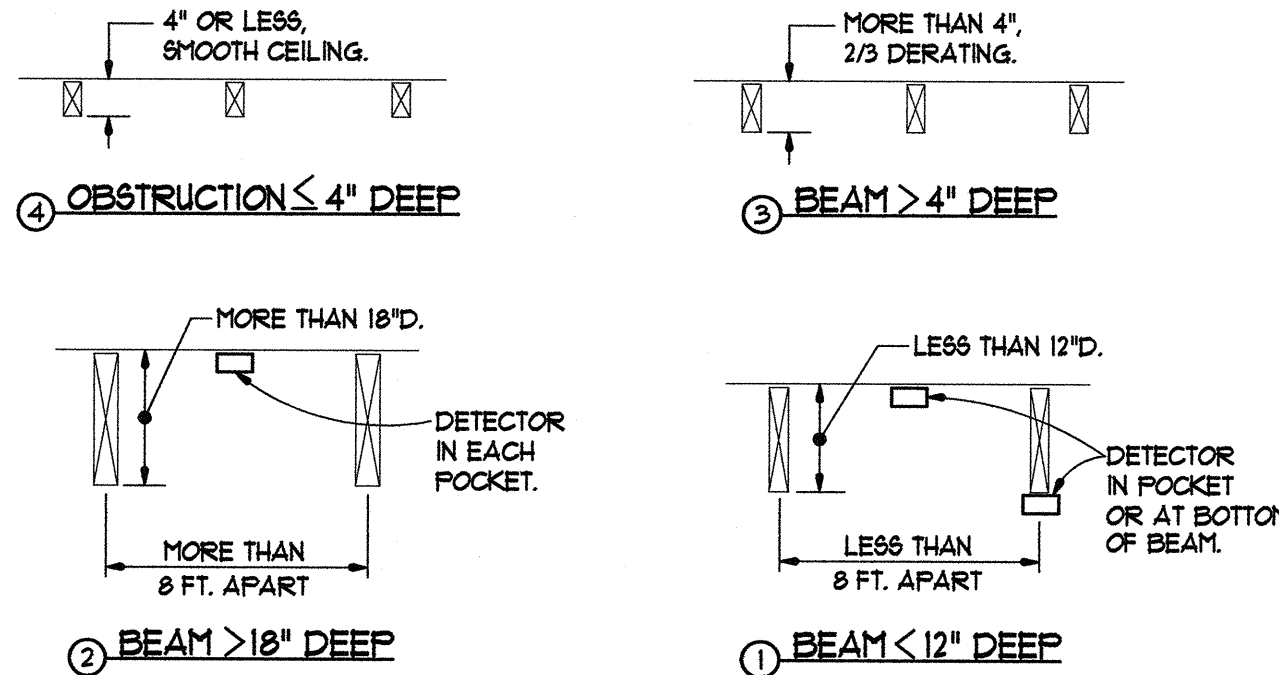
B
E5.4
FACP/ECC RISER DIAGRAM
NO SCALE



A
E5.4
TYPICAL - FIRE ALARM DEVICES WIRING DIAGRAMS
NO SCALE



D
E5.4
HEAT DETECTORS IN SOLID JOIST & BEAM CONSTRUCTION
NO SCALE



- NOTES:**
- SOLID JOIST CONSTRUCTION.
 - SPACING: THE DESIGN SPACING OF HEAT DETECTORS, WHEN MEASURED AT RIGHT ANGLES TO THE SOLID JOISTS, SHALL NOT EXCEED 90 PERCENT OF THE LISTED SPACING.
 - LOCATION: DETECTORS SHALL BE MOUNTED AT THE BOTTOM OF THE JOISTS.
 - BEAM CONSTRUCTION.
 - SPACING:
 - A CEILING SHALL BE TREATED AS A SMOOTH CEILING IF THE BEAMS PROJECT NO MORE THAN 4 IN. BELOW THE CEILING.
 - WHERE THE BEAMS PROJECT MORE THAN 4 IN. BELOW THE CEILING, THE SPACING OF SPOT-TYPE HEAT DETECTORS AT RIGHT ANGLES TO THE DIRECTION OF BEAM TRAVEL SHALL BE NOT MORE THAN TWO-THIRDS OF THE LISTED SPACING.
 - WHERE THE BEAMS PROJECT MORE THAN 18 IN. BELOW THE CEILING AND ARE MORE THAN 8 FT. ON CENTER, EACH BAY FORMED BY THE BEAMS SHALL BE TREATED AS A SEPARATE AREA.
 - LOCATION: WHERE BEAMS ARE LESS THAN 12 IN. IN DEPTH AND LESS THAN 8 FT. ON CENTER, DETECTORS SHALL BE PERMITTED TO BE INSTALLED ON THE BOTTOM OF BEAMS.
 - BEAM CONSTRUCTION CONSISTS OF SOLID MEMBERS PROJECTING DOWN FROM THE CEILING SURFACE MORE THAN 4" AND SPACED MORE THAN 36" O.C.
 - JOIST CONSTRUCTION CONSISTS OF SOLID MEMBERS PROJECTING DOWN FROM THE CEILING SURFACE MORE THAN 4" AND SPACED AT 36" OR LESS O.C.

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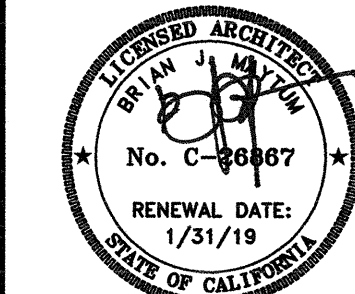
DIVISION OF THE STATE ARCHITECT
revision date: by:

HARRY A. YEE & ASSOCIATES, INC.
ELECTRICAL ENGINEERS
4920 FREEDPORT BLVD., SUITE D
SACRAMENTO, CALIFORNIA 95822
TEL: 916.454.5319
FAX: 916.454.4117
HYA Job #1602



CONSULTANT

nacht&lewis
600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtandlewis.com
916.329.4000



ARCHITECT

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JOB NO. Y1538.00

SHEET TITLE

Fire Alarm Details
Notes & Diagrams

SHEET NO.

E5.4

SHEET ____ OF ____ TOTAL

Dry Creek Joint Elementary School District

SHEET _____ OF _____ TOTAL

LOAD KVA: .

LOAD KYA: 52.9

LOAD KVA: 33.8

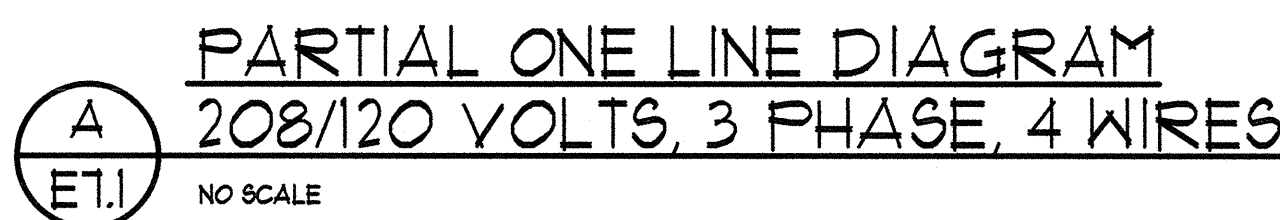
LOAD KYA: 30.4LOAD KYA:

LOAD KYA: 2.0

LOAD KVA: 94

LOAD KVA:

PANEL-DA, PROVIDE 3-600 KCMIL LUGS
PER PHASE.



208/120 VOLTS, 3 PHASE, 4 WIRES

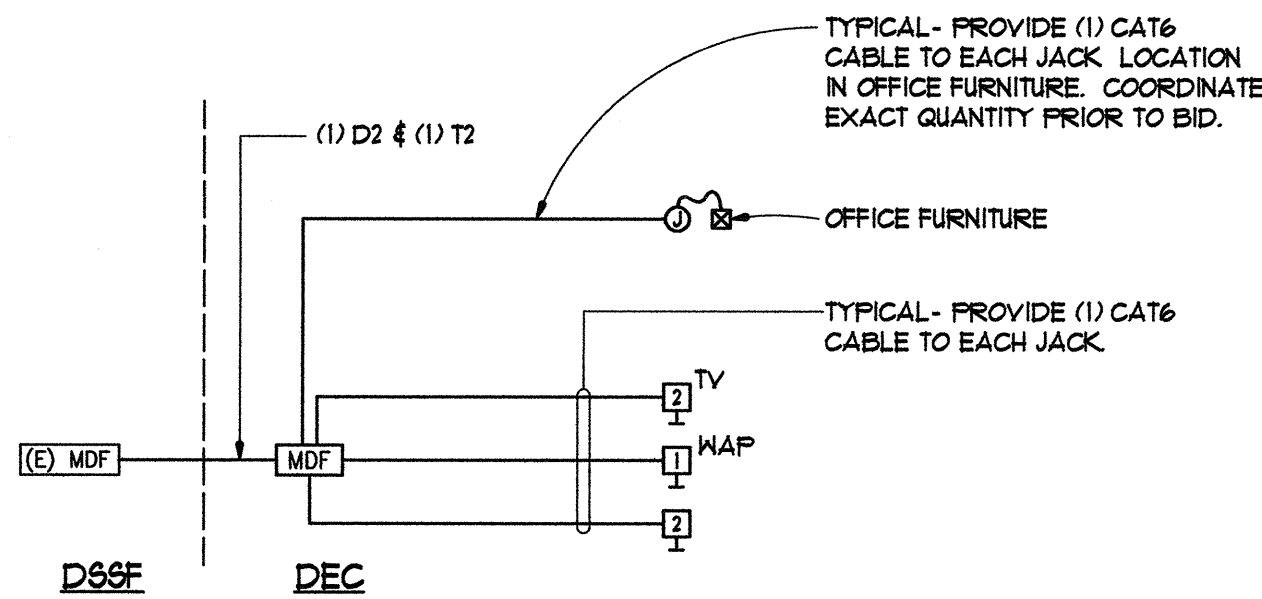
NO SCALE

ONE AND ONE-HALF INCH = ONE FOOT
ONE INCH = ONE FOOT
THREE-QUARTERS INCH = ONE FOOT
ONE-HALF INCH = ONE FOOT
ONE-QUARTER INCH = ONE FOOT
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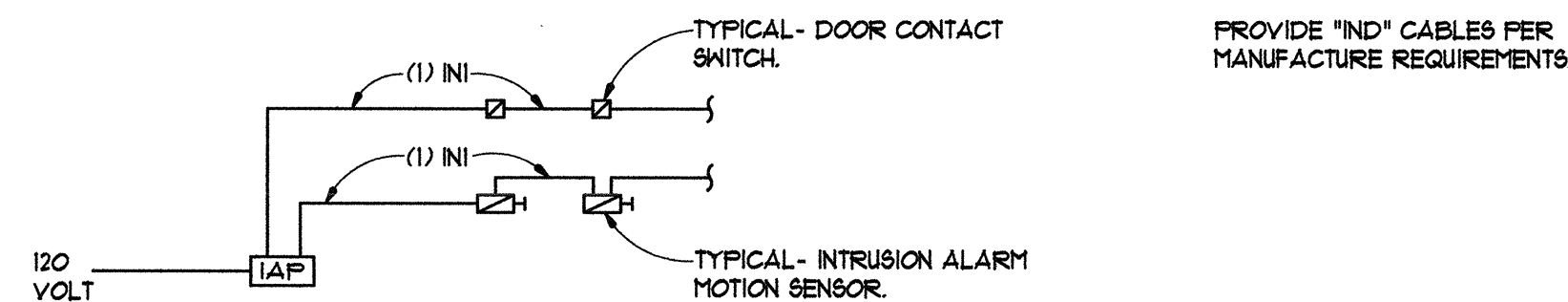
SIGNAL WIRE SCHEDULE					
DESIG.	DESCRIPTION	USE	SYSTEM	Q.D. (Inches)	AREA (sq. inches)
D1	4 PR #24, CAT-6, CM, BLUE	BUILDING	DATA		
D2	4 PR #24 CAT-6A, CM	BUILDING			
D3	FIBER OPTICS SEE SPEC'S.	SITE			
T1	4 PAIRS #24 CAT-6	BUILDING	TELEPHONE		
T2	25 PAIRS #24 CAT-6 NET-RATED	SITE			
IN1	CABLING PER MANUFACTURER REQUIREMENTS	BUILDING / SITE	INTRUSION		

NOTES:

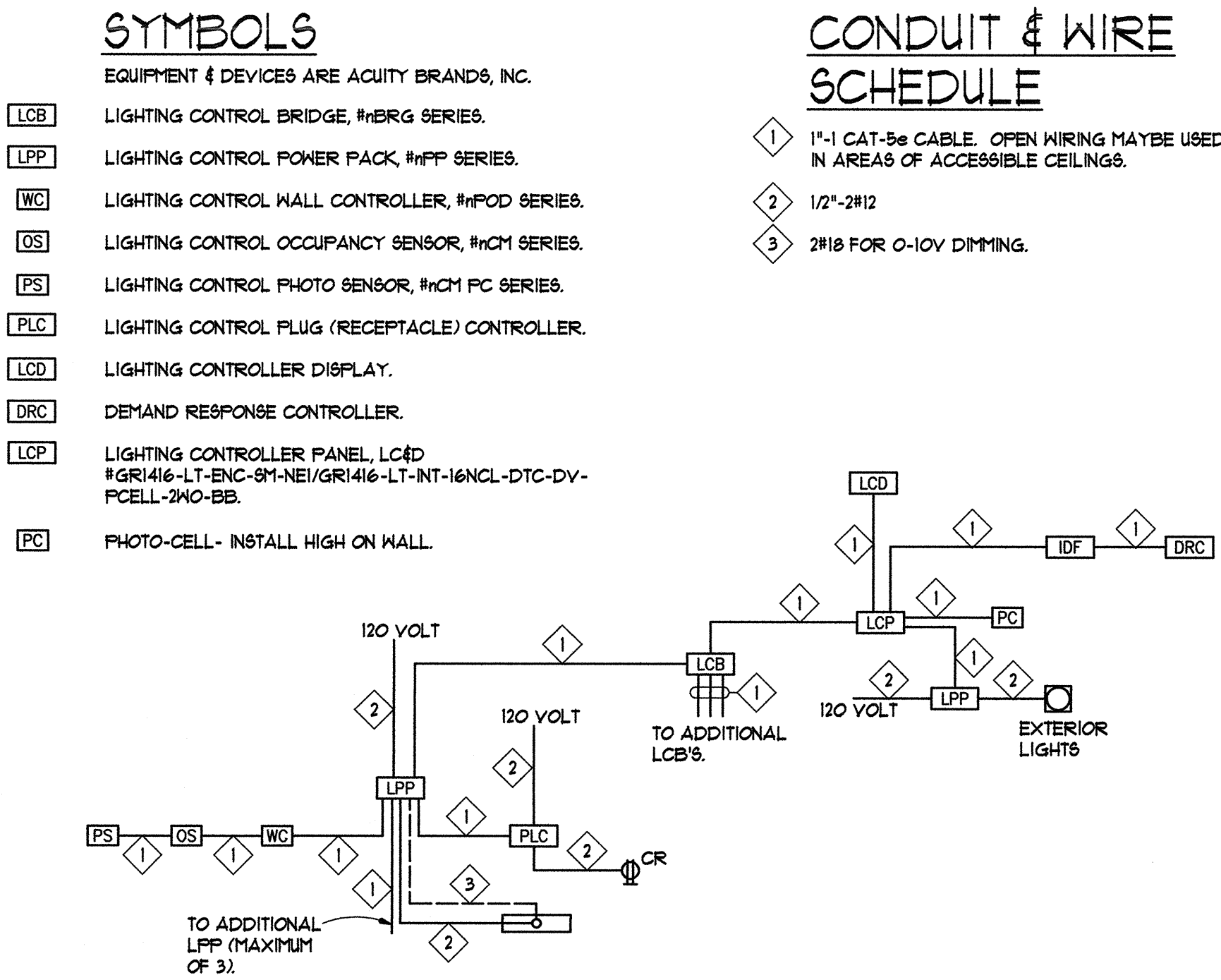
- ALL UNDERGROUND CABLES SHALL BE "NET RATED", MEET PERIN "AQUA-SEAL".
- NO UNDERGROUND SPLICES PERMITTED.



D
E8.1 DATA/COMMUNICATIONS RISER DIAGRAM
NO SCALE

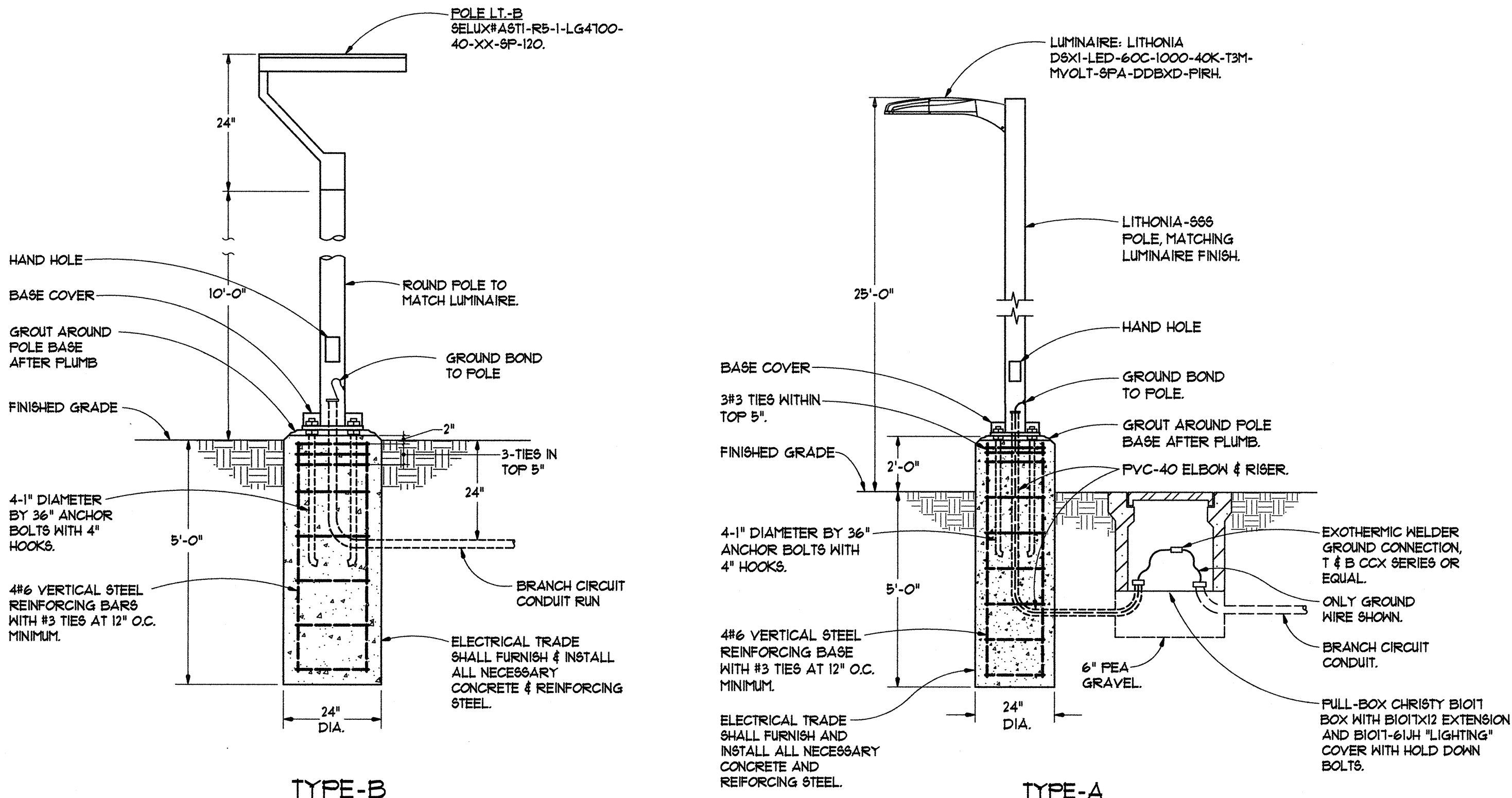


C
E8.1 INTRUSION ALARM WIRING DIAGRAM
NO SCALE



B
E8.1 TYPICAL - LIGHTING CONTROL SYSTEM WIRING DIAGRAM
NO SCALE

LIGHT FIXTURE SCHEDULE						
SYM	MANUFACTURER AND CATALOG NUMBER	LAMPS	MOUNTING	DESCRIPTION	REMARKS	WATTS
A12	FINELITE 612LEDID-DCO-12-4E-H-8-35-10U90D-120- SC-XX-CE-XX	LED (5000 LUMENS/4 FT.)	SUSPENDED 4'-0"-0"	12' DIRECT/INDIRECT LINEAR		140
A24	FINELITE 612LEDID-DCO-24-4E-H-8-35-10U90D-120- SC-XX-CE-XX			24' DIRECT/INDIRECT LINEAR		281
A28	FINELITE 612LEDID-DCO-28-4E-H-8-35-10U90D-120- SC-XX-CE-XX			28' DIRECT/INDIRECT LINEAR		328
A36	FINELITE 612LEDID-DCO-36-4E-H-8-35-10U90D-120- SC-XX-CE-XX			36' DIRECT/INDIRECT LINEAR		421
B6	FINELITE HP-28T-6-V-8-35-BG-120V-XX-SC	LED (3215 LUMENS/4 FT.)	SURFACE	6' DIRECT LINEAR		55
B8	FINELITE HP-28T-8-V-8-35-BG-120V-XX-SC			8' DIRECT LINEAR		74
B12	FINELITE HP-28T-12-V-8-35-BG-120V-XX-SC			12' DIRECT LINEAR		112
C1	FINELITE HP-28T-1-V-8-35-BG-120V-XX-SC	LED (3215 LUMENS/4 FT.)	SUSPENDED 4'-0"-0"	1' DIRECT LINEAR		64
C8	FINELITE HP-28T-8-V-8-35-BG-120V-XX-SC			8' DIRECT LINEAR		74
C10	FINELITE HP-28T-10-8-35-BG-120V-XX-SC			10' DIRECT LINEAR		92
D	FINELITE HPR-LED-A-2X4-DCO-H-8-35- 120V-SC-XX	LED (5416 LUMENS)	RECESSED	2' x 4' TROFFER		41
E	FINELITE HPR-LED-A-2X2-DCO-S-8-35- 120V-SC-XX	LED (3391 LUMENS)		2' x 2' TROFFER		31
F	HE WILLIAMS AT3-2-4-L-10-D-D-UNV	LED (1000 LUMENS)	SURFACE	2' x 4' METAL SIDED		69
G8	FINELITE MU-O-8-LED-35-SC-120V-WH-FE	LED (349 LUMENS/4 FT.)	WALL	8' DIRECT LINEAR		36
G12	FINELITE MU-O-12-LED-35-SC-120V-WH-FE			12' DIRECT LINEAR		54
G24	FINELITE MU-O-24-LED-35-SC-120V-WH-FE			24' DIRECT LINEAR		108
H6	FINELITE HPN-LED-6-FR-B-8-35-120V-SC-XX	LED (2010 LUMENS/4 FT.)	RECESSED	6" WALL WASHER		28
H8	FINELITE HPN-LED-8-FR-B-8-35-120V-SC-XX			8" WALL WASHER		31
H16	FINELITE HPN-LED-16-FR-B-8-35-120V-SC-XX			16" WALL WASHER		74
K	HE WILLIAMS T1-4-L95-AF-120V	LED (3500 LUMENS)	SURFACE	1' x 4' WRAP AROUND		53
L	HE WILLIAMS L60-L20C-G3-W-120V	LED (2000 LUMENS)	RECESSED	INTERIOR CAN WITH LENS		26
M	GOTHAM EVO-VR-40-20-XX-L86-T13-MVOLT-EZ	LED (2000 LUMENS)				21
N	LITHONIA DSXWLED-10C-100-40K-T3H-MVOLT-XX	LED (2151 LUMENS)	WALL 4'-0"			21
P	THS 8WHL-15LED-35K-120-RAL-G3	LED	COLUMN 4'-6"			15
E81	McPHILBEN 65x6N124N12	2-12W-T8R16	WALL 4'-0"-0"	EMERGENCY BATTERY UNIT WITH 2 FIXTURES.		24
E82	McPHILBEN 65x6N184N18CH	2-18W-T8R16	WALL 4'-0"-0"	EMERGENCY BATTERY UNIT WITH 2 FIXTURES.	WITH HEATER.	18



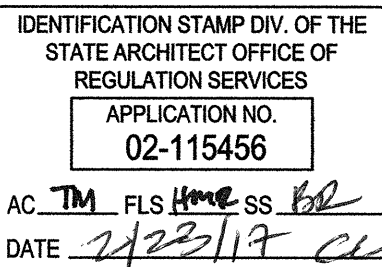
A
E8.1 POLE LIGHT DETAILS
NO SCALE

Dry Creek Joint Elementary School District
District Education Center

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Dry Creek Joint Elementary School District

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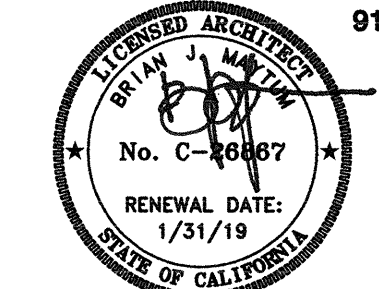
HARRY A. YEE & ASSOCIATES, INC.
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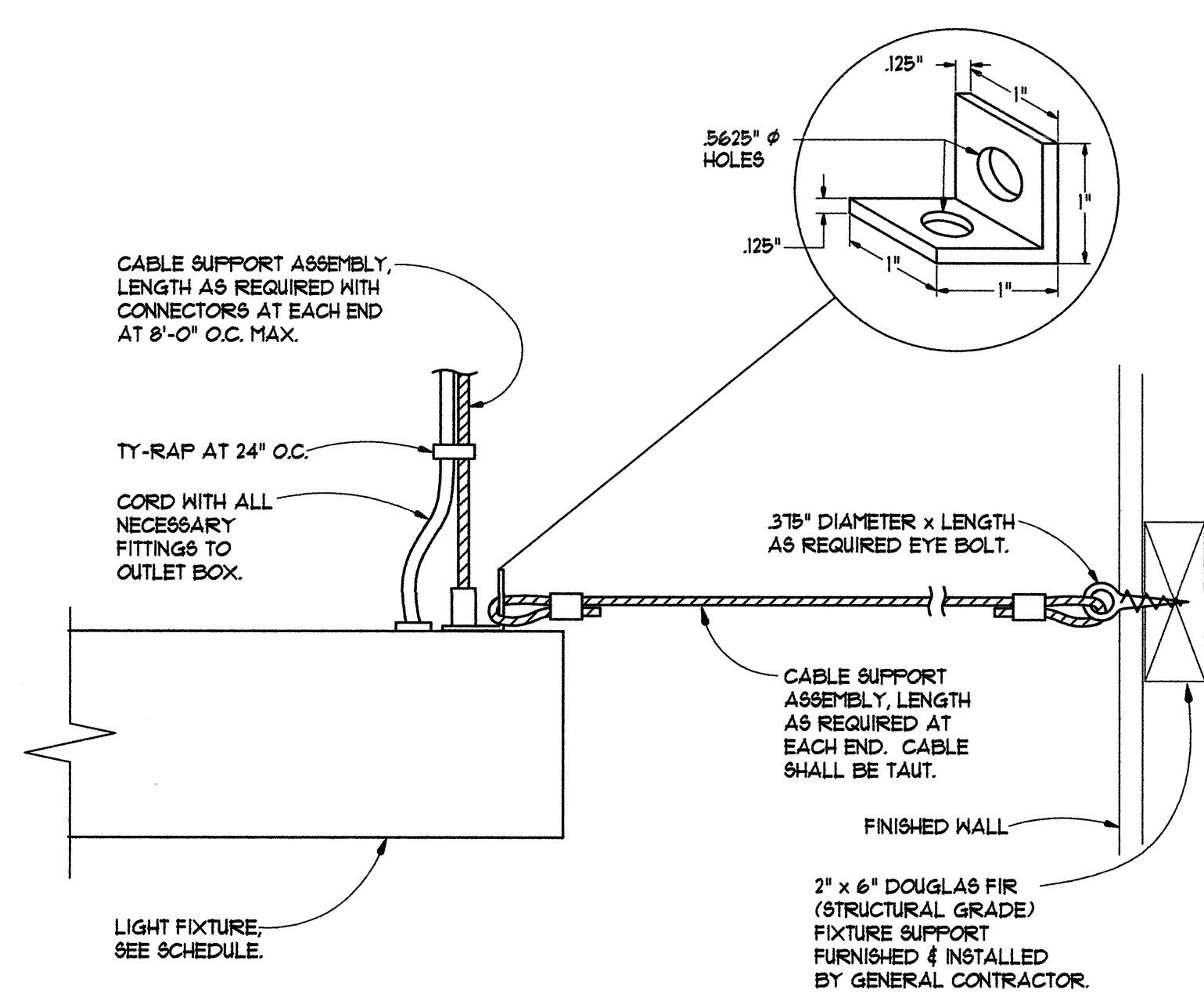
SHEET TITLE

Light Fixture
Schedules,
Notes & Details

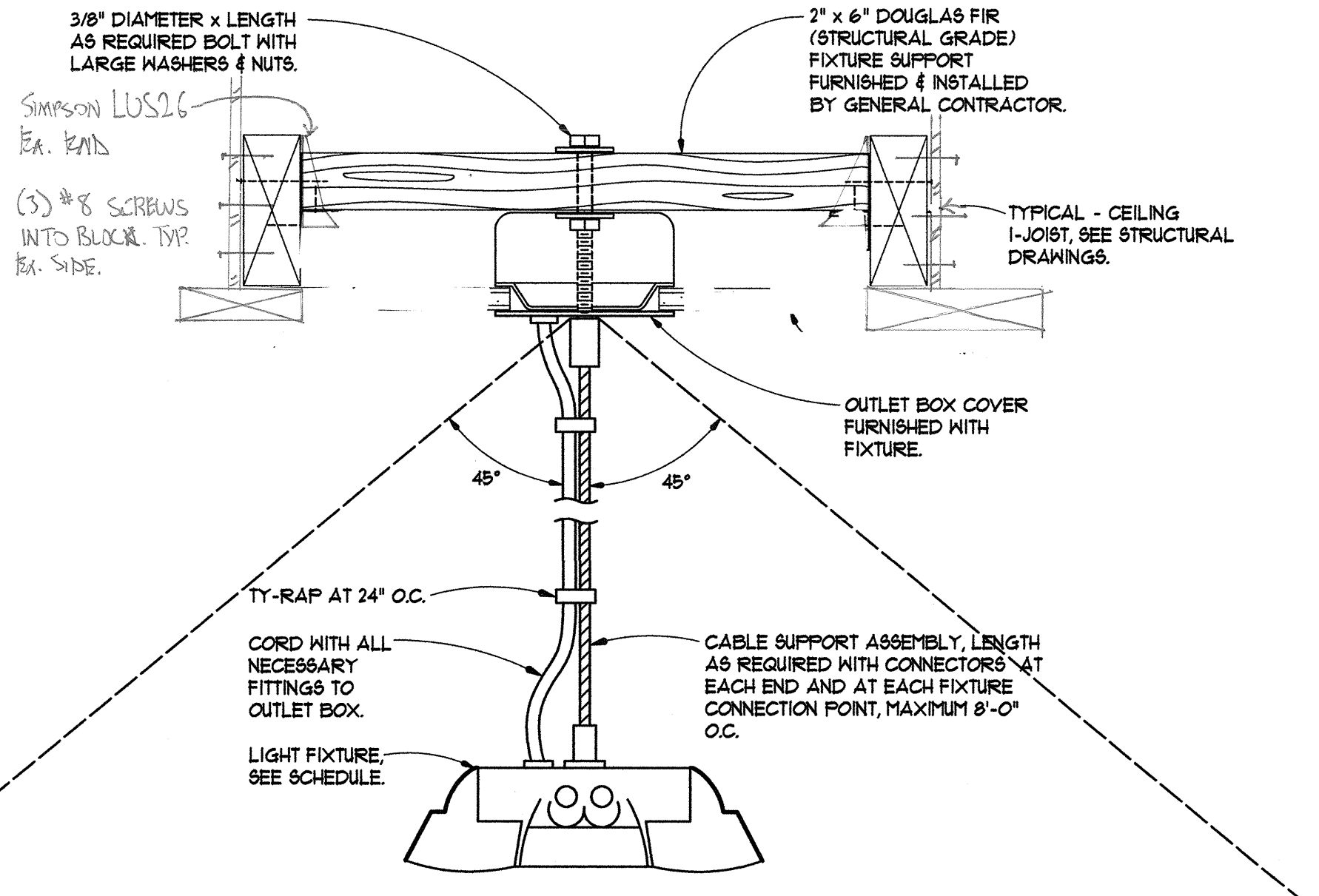
SHEET NO.

E8.1

SHEET ____ OF ____ TOTAL



NO OBSTRUCTIONS
ALLOWED OR BRACE -
SUSPENDED LIGHT
FIXTURE TO STOP
SWINGING.



A LIGHT FIXTURE MOUNTING DETAILS
E8.2 NO SCALE

- | <u>TYPE OF CEILING</u> | <u>LUMINAIRE SIZE</u> | <u>LUMINAIRE WEIGHT</u> | <u>WIRE HANGERS</u> |
|------------------------|-----------------------|-------------------------|---------------------|
| HEAVY DUTY | 2 x 4 OR LESS | < 56 LBS. | 2#12 GA. (SLACK) |
| HEAVY DUTY | 4 x 4 | < 56 LBS. | 4#12 GA. (SLACK) |
| ALL | ALL | > 56 LBS. | 4#12 GA. (TAUT) |
| INTERMEDIATE | ALL | ALL | 4#12 GA. (TAUT) |

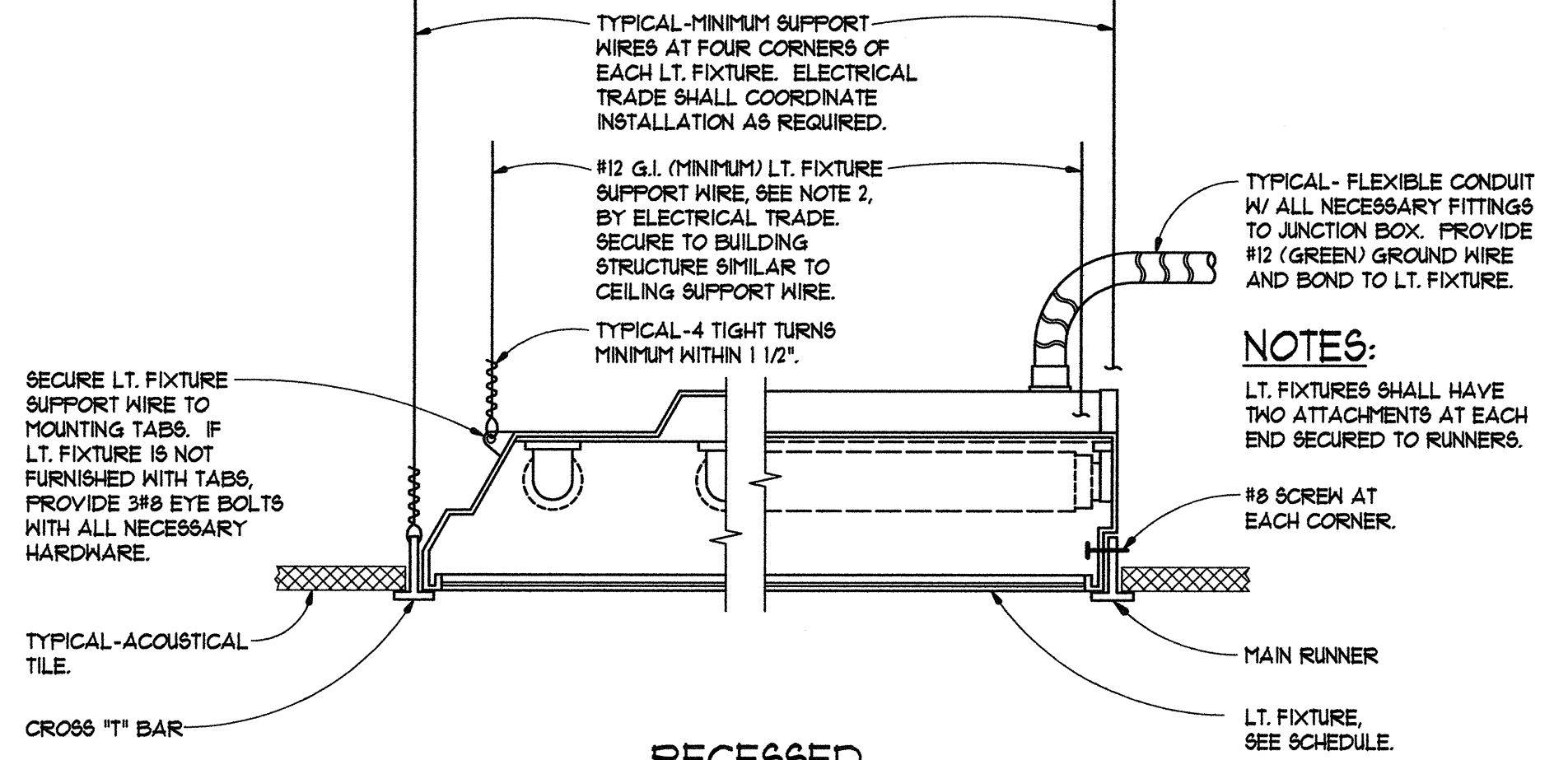
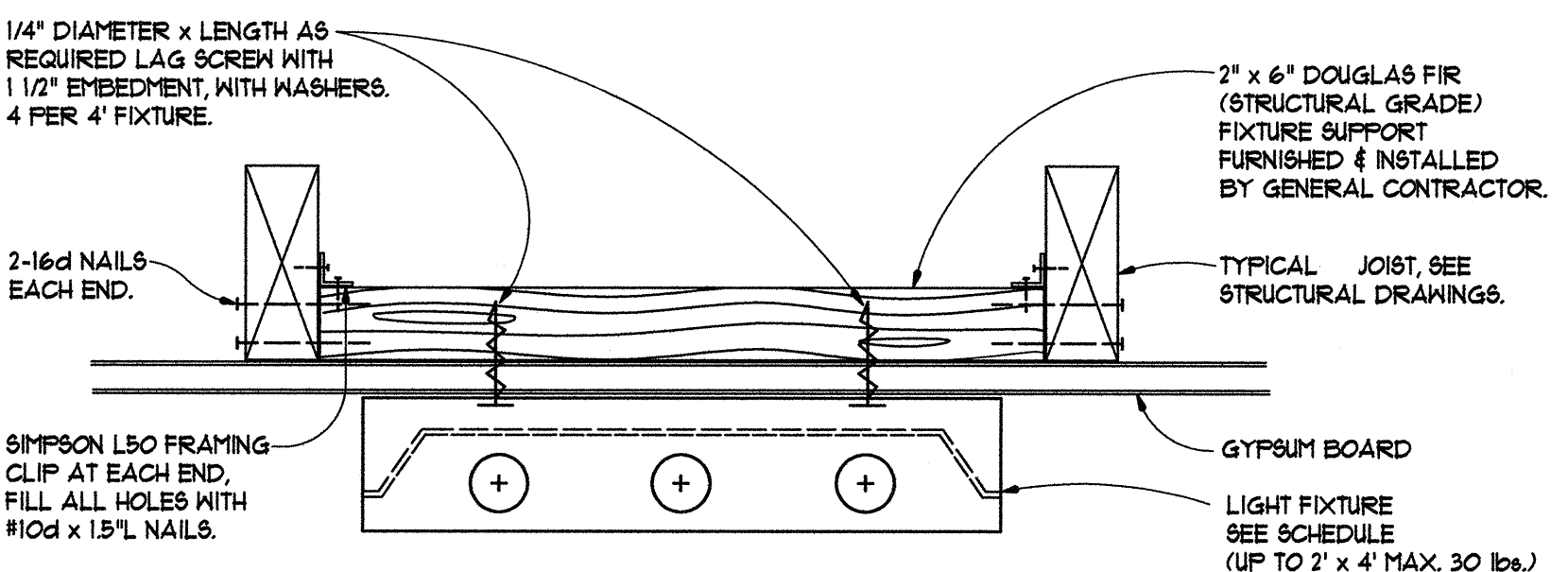


Diagram illustrating a typical support member (N.I.E.S.) for a recessed "down" light fixture. The fixture is mounted within a rectangular opening in a ceiling. The installation includes:

- TYPICAL - SUPPORT MEMBER (N.I.E.S.)**: The structural support for the fixture.
- RECESSED "DOWN" LIGHT FIXTURE (MAXIMUM 15 lbs.)**: The light fixture itself, mounted to the support member.
- ADJUSTABLE "C" BRACKETS WITH 5-6 BAR HANGERS**: Used to secure the fixture to the support member.
- FLEXIBLE CONDUIT**: Used for wiring the fixture.
- TYPICAL - 2-1/2" NAILS AT EACH END**: Used to secure the support member to the ceiling.

SURFACE ON T-BAR



SURFACE ON GYP BOARD

DAVID K. YU
REGISTERED PROFESSIONAL ENGINEER
No. E14987
Exp. 6/30/17
ELECTRICAL
STATE OF CALIFORNIA

nacht&lewis
600 Q Street, Suite 100
Sacramento, CA 95811
www.nachtlewis.com
916.329.4000

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